MIL-M-28787/62C(EC)
22 January 1979
SUPERSEDING
MIL-M-28787/62B(EC)
27 March 1978

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COMPARATOR, ANALOG, KEY CODE CFA

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic analog comparator module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table II.

Name	Comparator, analog
Family	Analog
Key code	CFA
Size	1A
Weight	35 grams maximum
Failure rate	0.444 failures/10^6 hours maximum
Environment	Class I as specified in MIL-STD-1389
Maximum power dissipation	
Without pullup resistors connected.	
+16.5 V supply	710 mW
-16.5 V supply	595 mW
Without pullup resistors connected.	
+5.5 V supply (to pin 1 or 21)	200 mW
With pullup resistors connected to +5.5 V and the output of the comparators.	
+16.5 V supply	710 mW
-16.5 V supply	595 mW
+5.5 V supply (to pin 2)	220 mW
Supply voltage (A, B, or C but not simultaneously).	
A	+15 Vdc ±10 percent
A	-15 Vdc ±10 percent
A	+ 5 Vdc ±10 percent (optional)
B	+ 5 Vdc ±10 percent
C	+30 Vdc maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN ELEX 5043, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1 Comparator description.

Comparator type	Open collector output
Input offset voltage	4.0 mV maximum
Input offset current	15.0 nA maximum
Input bias current	150 nA maximum
Low-level output voltage	
I_O = 5.5 mA	0.4 V maximum
I_O = 50 mA	1.5 V maximum
Common mode voltage range	Between ($-V_{CC}$ +0.4 V) and ($+V_{CC}$ -1.3 V)
Output leakage current	
V_{OUT} = 35 V	100 nA maximum
Response time, t_{RLH}	400 ns maximum
Response time, t_{RHL}	400 ns maximum

1.2.2 Pullup resistor description.

Pullup resistor	1 kΩ ±15%, 60 mW

1.3 Module function. This module contains six independent open collector analog comparators and six 1 kΩ resistors to be used as pullup resistors for comparator outputs.

1.3.1 Input to output relationship Each comparator has the input to output relationship of table I.

1.3.2 Common mode voltage range. The common mode voltage range is between ($-V_{CC}$ +0.4 V) and ($+V_{CC}$ - 1.3 V) when the input common mode voltage exceeds the common mode voltage range, the comparator output may latch at a high level. Test condition R guarantees correct module operation with a minimum common mode voltage of ($+V_{CC}$ - 1.3 V).

TABLE I. Input to output relationship.

Input voltage	Output
+ Terminal voltage > - terminal voltage	High level
+ Terminal voltage < - terminal voltage	Low level

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

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STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.444 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirments specified in table II and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be CFA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

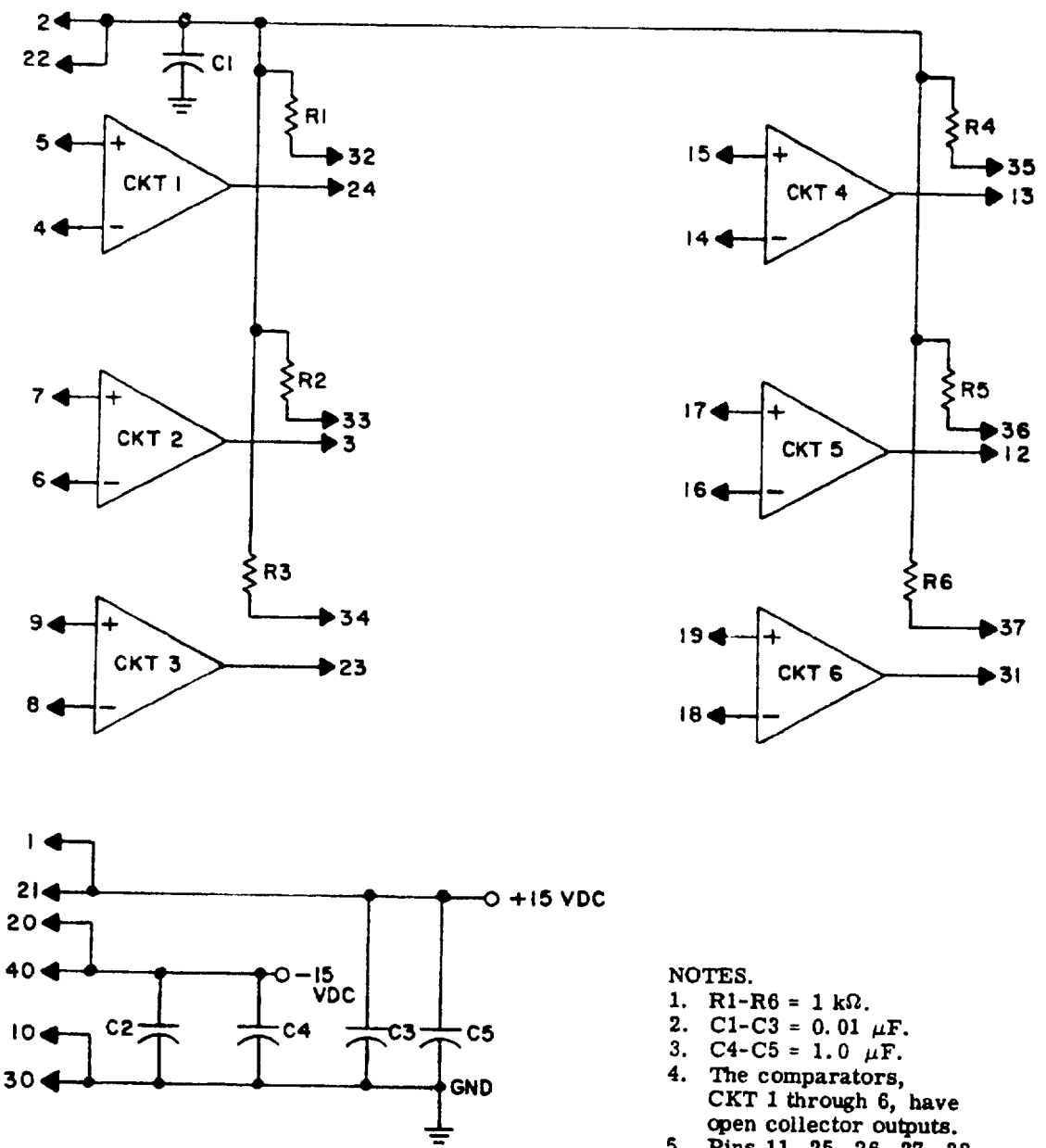
3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/62, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

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NOTES.
1. R1-R6 = 1 kΩ.
2. C1-C3 = 0.01 μF.
3. C4-C5 = 1.0 μF.
4. The comparators, CKT 1 through 6, have open collector outputs.
5. Pins 11, 25, 26, 27, 28, 29, 38, and 39 are unused.

FIGURE 1 CFA module function diagram, analog comparator

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TABLE II. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A9	Isolation	III	X	10.0	∞	1.0	∞	10.0	∞	1.0	∞	MΩ
A10	Isolation, frame	III	S	10.0	∞	1.0	∞	10.0	∞	1.0	∞	MΩ
B1-B6	Stress	IV	S	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	V
C1-C12	Offset voltage	V	X	-3.0	3.0	-3.0	3.0	-4.0	4.0	-4.0	4.0	mV
D1-D24	Input bias current	VI	X	0.0	100	0.0	100	0.0	150	0.0	150	nA
E1-E12	Input offset current	VII	X	-10	10	-10	10	-15	15	-15	15	nA
F1-F12	Low-level output voltage	VIII	X	0.01	0.40	0.01	0.40	0.01	0.40	0.01	0.40	V
F13-F18	Low-level output voltage	VIII	S	0.1	1.5	0.1	1.5	0.1	1.5	0.1	1.5	V
F19-F24	Low-level output voltage	VIII	S	0.01	0.40	0.01	0.40	0.01	0.40	0.01	0.40	V
G1-G6	Output leakage current	IX	X	0.0	50.0	0.0	50.0	0.0	100.0	0.0	100.0	nA
H1-H6	Resistance	X	X	925	1075	888	1113	888	1113	850	1150	Ω
J1-J4	Continuity	X	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ω
K1,K3	Power supply current (+16.5 V)	XI	X	1.0	38.0	1.0	38.0	1.0	43.0	1.0	43.0	mA
K2,K4	Power supply current (-16.5 V)	XI	X	1.0	31.5	1.0	31.5	1.0	36.0	1.0	36.0	mA
K5	Power supply current (single +5.5 V)	XI	X	1.0	31.5	1.0	31.5	1.0	36.0	1.0	36.0	mA
L1-L6	Response time (t_{RLH})	XII	S	30	400	30	400	30	400	30	400	ns
L7-L12	Response time (t_{RHL})	XII	S	30	400	30	400	30	400	30	400	ns
M1,M2	Power supply filter capacitance (±15 V)	XIII	S	0.75	1.25	0.75	1.25	0.75	1.25	0.75	1.25	µF
M3	Power supply filter capacitance (pin 2)	XIII	S	0.008	0.012	0.008	0.012	0.008	0.012	0.008	0.012	µF
N1-N6	Output crosstalk, low level	XIV	Q	0	500	0	500	0	500	0	500	mV
N7-N12	Output crosstalk, high level	XIV	Q	0	400	0	400	0	400	0	400	mV
P1-P6	Response time (t_{RLH}), V_{CC} = +30 V	XV	Q	30	400	30	400	30	400	30	400	ns

See notes at end of table.

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TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° +5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P7-P12	Response time (t_{RHL}), V_{CC}= +30 V	XV	Q	30	400	30	400	30	400	30	400	ns
R1-R12	Common mode voltage, high level	XVI	X	3.100	4.545	3.100	4.545	3.100	4.545	3.100	4.545	V

NOTES

1. (X) parameters shall be 100 percent tested during acceptance tests.
2. (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C
3. (Q) parameters shall be tested during qualification testing only.

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3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed two years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below.

Equipment	MIL-STD-1665 Designator	No. required (maximum per test figure)
DC power supply	Item P07	5
Megohm bridge	Item R06	1
DC voltmeter	Item V01	1
Digital multifunction meter	Item M02	1
DC voltmeter	Item V07	1
DC power supply	Item P01	1
DC ammeter	Item X03	1
DC ammeter	Item X01	2
Pulse generator	Item J01	1
Oscilloscope	Item D05	1
Wideband probe	Item Y01	2
Capacitance bridge	Item C01	1
Oscilloscope	Item D01	1
Resistors		
49.9Ω ±0.1%, 1/8 W		2
49.9 kΩ ±0.1%, 1/8 W		2
100 kΩ ±1%, 1/8 W		3
10.0 kΩ ±1%, 1/8 W		8
500Ω, 0.5 W potentiometer		1
27.4Ω ±1%, 1/8 W		1
23.7Ω ±1%, 1/8 W		1
1.50 kΩ ±1%, 1/8 W		1
5.30 kΩ ±1%, 1/8 W		1
274Ω ±1%, 1 W		1
31.6Ω ±1%, 1/8 W		1
2.0 kΩ ±1%, 1/8 W		1
20.5 kΩ ±1%, 1/8 W		1
100Ω ±1%, 1/8 W		1
249 kΩ ±1%, 1/8 W		1
5.36 kΩ ±1%, 1/8 W		2
100Ω, 3/4 W potentiometer		1
51Ω ±5%, 1/2 W		1

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Resistors - Continued.

25.5Ω ±1%, 1/8 W	1
49.9 kΩ ±1%, 1/8 W	2
2.49 kΩ ±1%, 1/8 W	2
49.9Ω ±1%, 1/8 W	2
50Ω ±5%, 1 W	1
2.7 kΩ ±1%, 1/8 W	1
1.3 kΩ ±1%, 1/8 W	1

Capacitor

0.01 µF	1
0.001 µF ±10%	2

Operational amplifier

741, or equivalent	1

Miscellaneous

Switch, SPST	2

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table II.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program (SEMP) Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through R shall be performed in any order with the following exceptions. If test conditions A and B are part of a sequence of tests, the order of testing shall be as follows A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through R in table II shall be performed in the order specified in 4.4.2 using the initial limits at 25° +5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A and C through R shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table II which are marked with an (X) in the test type column, and test condition M, shall be performed using the initial limits at 25° ±5°C ambient.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests A and C through R shall be performed in the order specified in 4,4.2 using the operating temperature end-of-life requirements of table II. Following the return of the module temperature to 25°C, electrical tests A and C through R shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table II.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life.

Preparing activity
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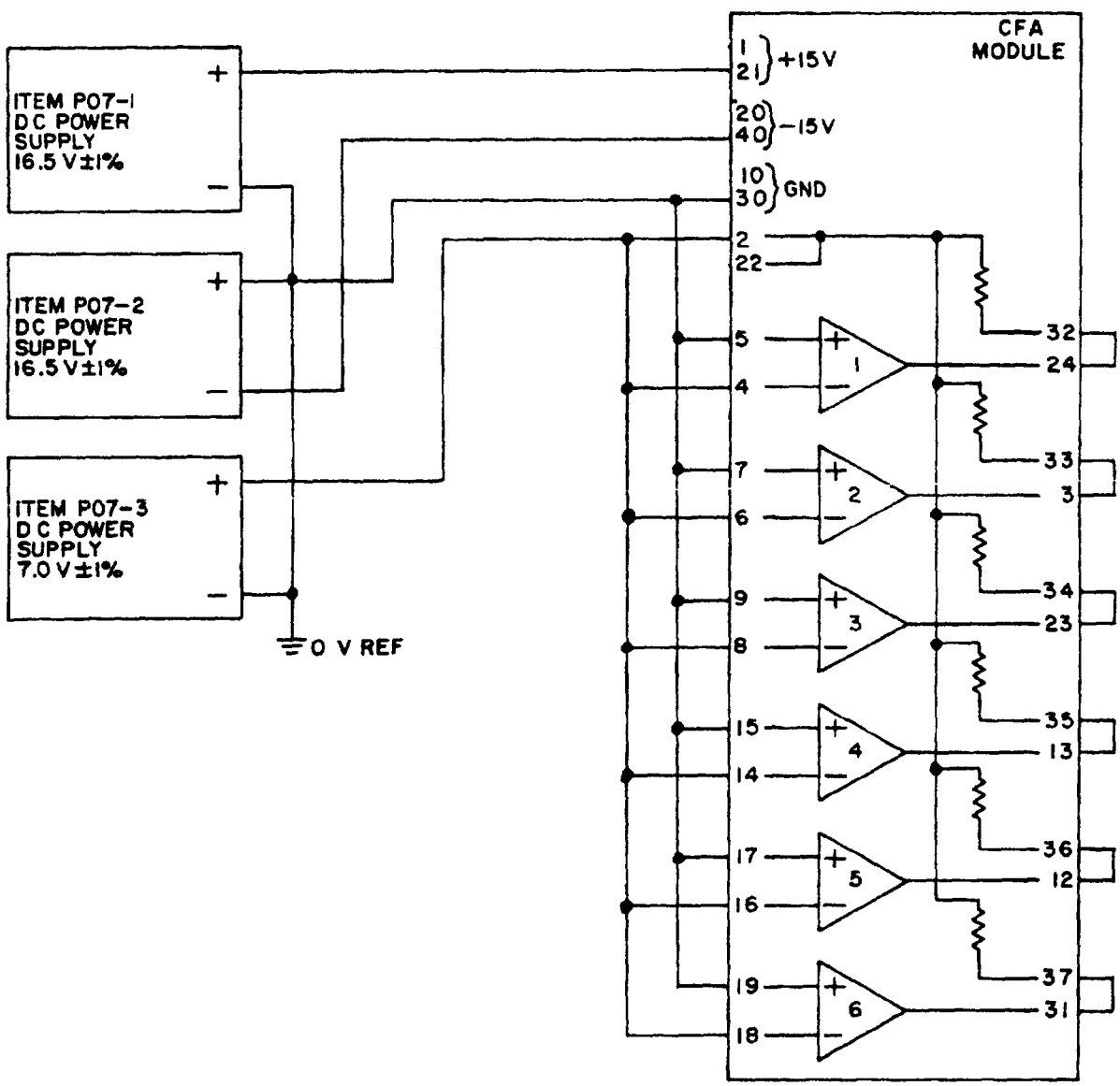


FIGURE 2 Operating temperature, component temperature, and life test circuit.

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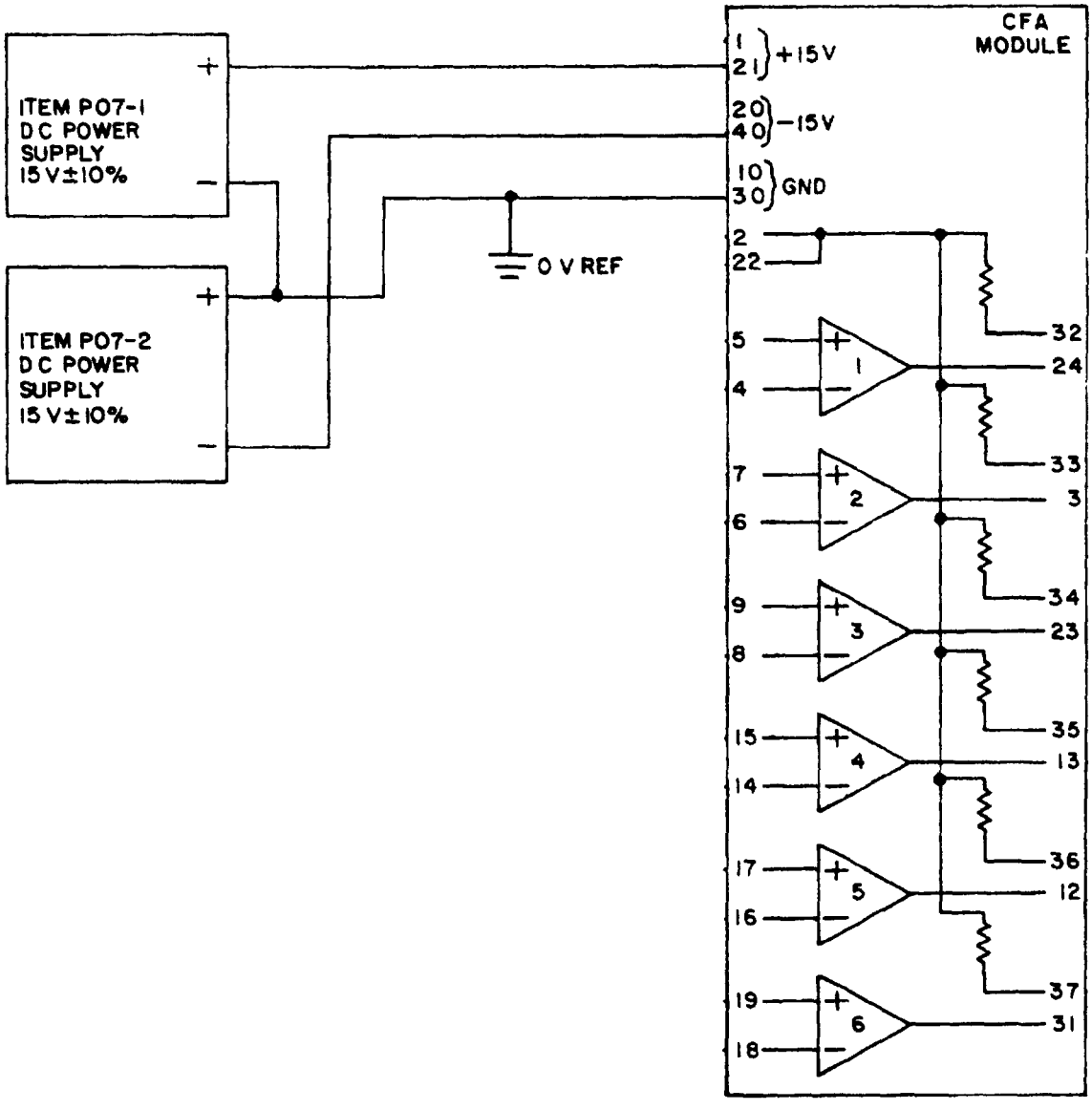


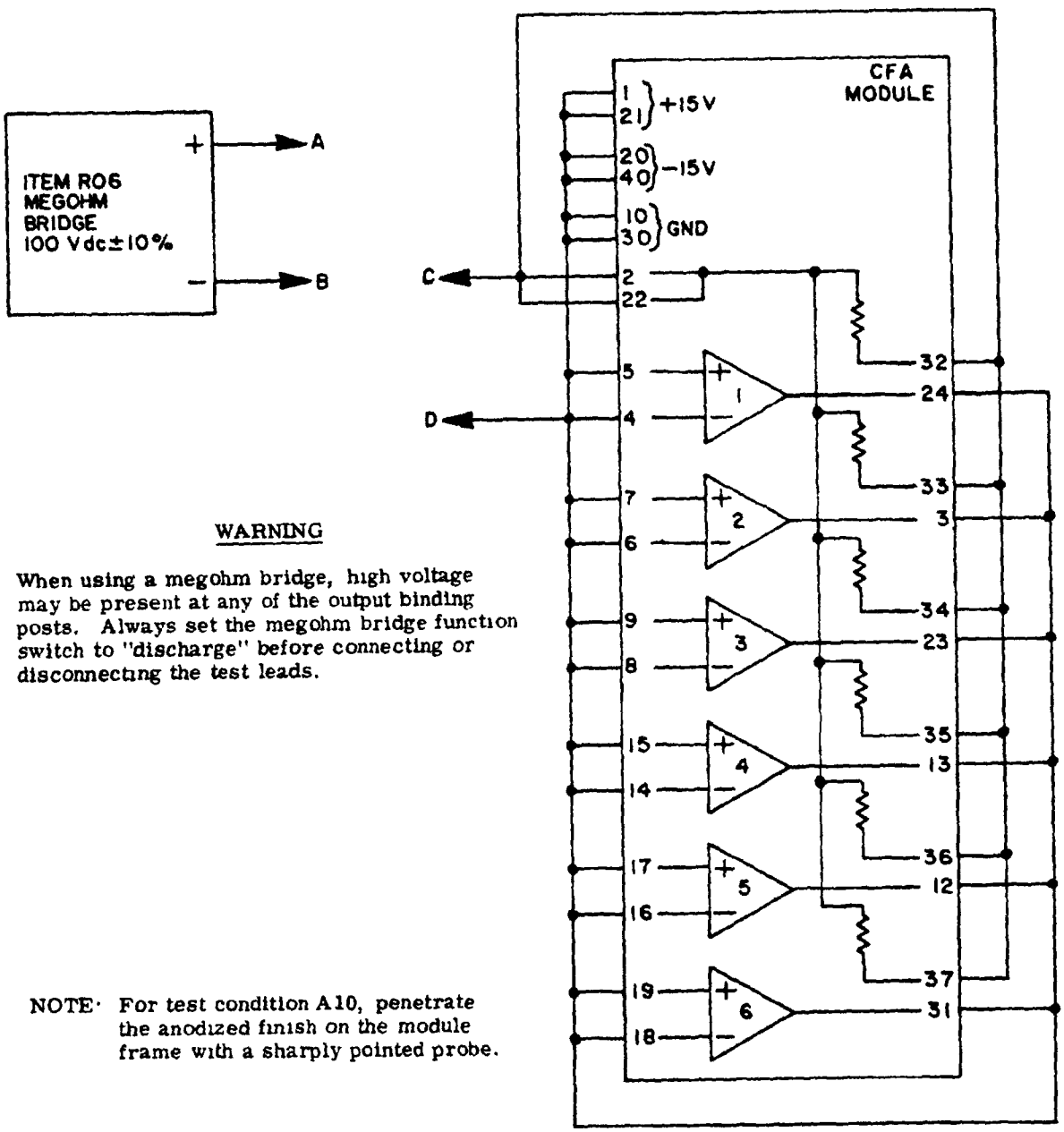
FIGURE 3. Vibration test circuit.

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TABLE III. Isolation test

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, circuit to circuit	C	D	10	∞	MΩ	4
A2	Isolation, used to unused pins	C,D	11,25,26,27,28,29,38,39	↓	↓	↓	↓
A3	Isolation, unused to unused pins	11	25,26,27,28,29,38,39				
A4	↓	25	26,27,28,29,38,39				
A5		26	27,28,29,38,39				
A6		27	28,29,38,39				
A7		28	29,38,39				
A8		29	38,39				
A9	Isolation, unused to unused pins	38	39				
A10	Isolation, frame	C,D	Frame	10	∞	MΩ	4

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A10, penetrate the anodized finish on the module frame with a sharply pointed probe.

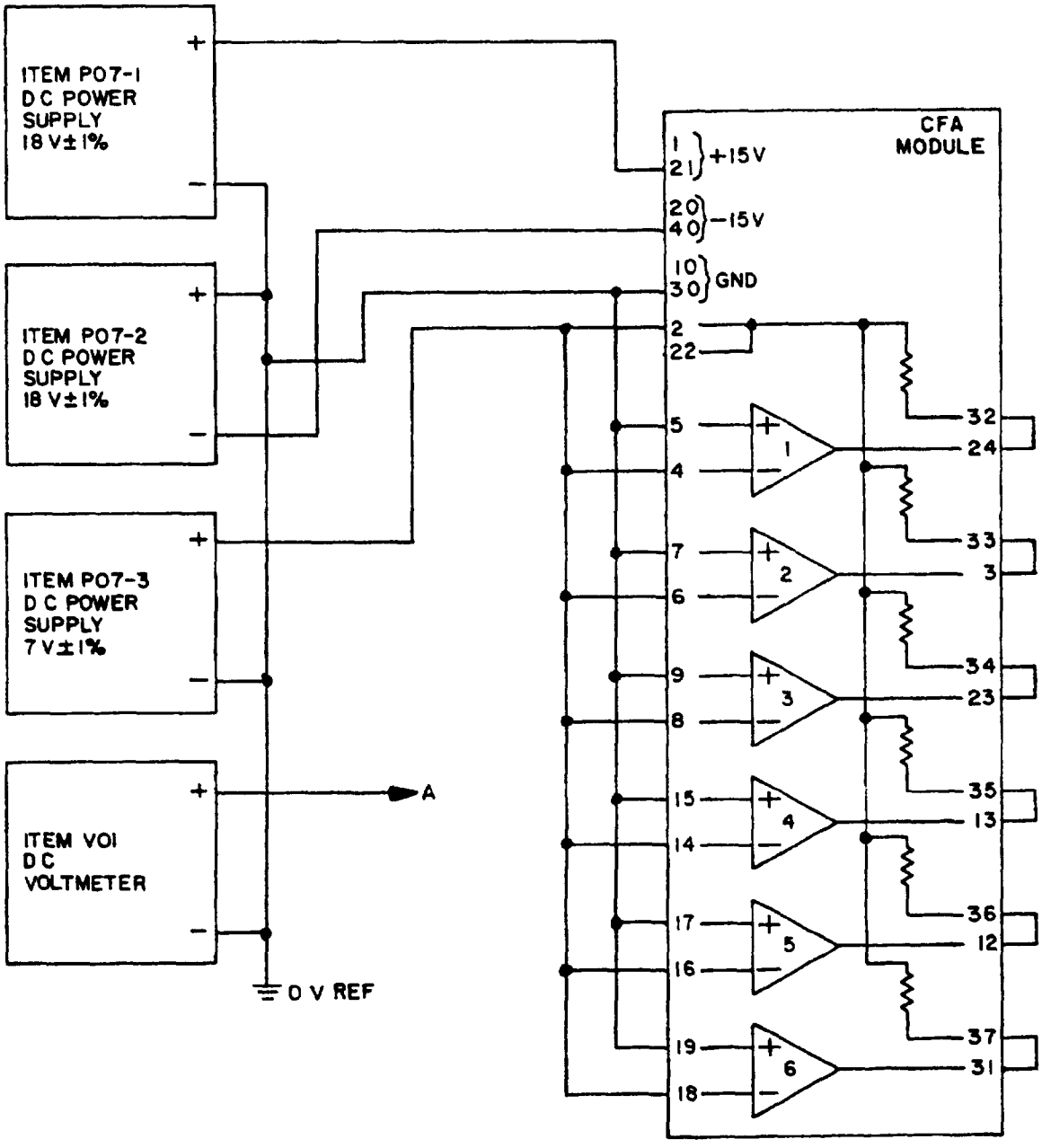
FIGURE 4. Isolation test circuit for table III.

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TABLE IV. Stress test

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
B1	Stress	24	0.01	0.8	V	5
B2	↓	3	↓	↓	↓	↓
B3		23				
B4		13				
B5		12				
B6	Stress	31	0.01	0.8	V	5

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NOTE Make the required connections and apply power to the module for 1 minute before making measurements.

FIGURE 5. Stress test circuit for table IV.

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TABLE V. Offset voltage test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
C1	Offset voltage (+ 15 V)	5	4	24	- 3.0	+ 3.0	mV	6
C2	↓	7	6	3	↓	↓	↓	↓
C3	↓	9	8	23	↓	↓	↓	↓
C4	↓	15	14	13	↓	↓	↓	↓
C5	↓	17	16	12	↓	↓	↓	↓
C6	Offset voltage (± 15 V)	19	18	31	↓	↓	↓	6
C7	Offset voltage (+ 5 V)	5	4	24	↓	↓	↓	7
C8	↓	7	6	3	↓	↓	↓	↓
C9	↓	9	8	23	↓	↓	↓	↓
C10	↓	15	14	13	↓	↓	↓	↓
C11	↓	17	16	12	↓	↓	↓	↓
C12	Offset voltage (+ 5 V)	19	18	31	- 3 0	+ 3.0	mV	7

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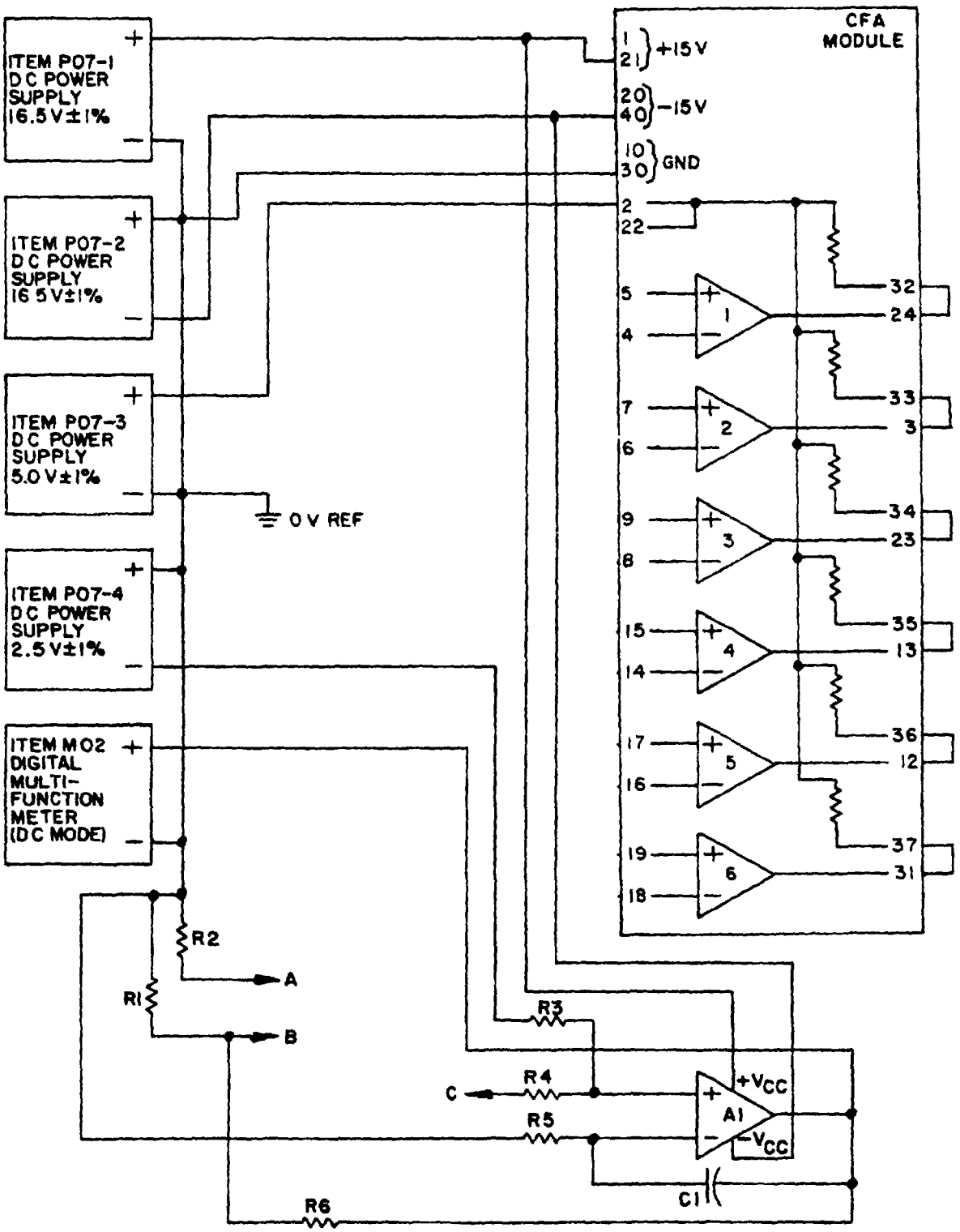


FIGURE 6. Offset voltage (+15 V) test circuit for table V (test conditions C1 through C6).

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NOTES:
1. R1 = R2 = 49.9Ω ±0.1%, 1/8 W.
2. R3 = R4 = R5 = 100 kΩ ±1%, 1/8 W.
3. R6 = 49.9 kΩ ±0.1%, 1/8 W.
4. C1 = 0.01 μF.
5. A1 = Operational amplifier 741, or equivalent.
6. For each test condition, divide the M02 reading by 1,000. The resultant shall be within the table V limits.

FIGURE 6. Offset voltage (+15 V) test circuit for table V (test conditions C1 through C6) - Continued.

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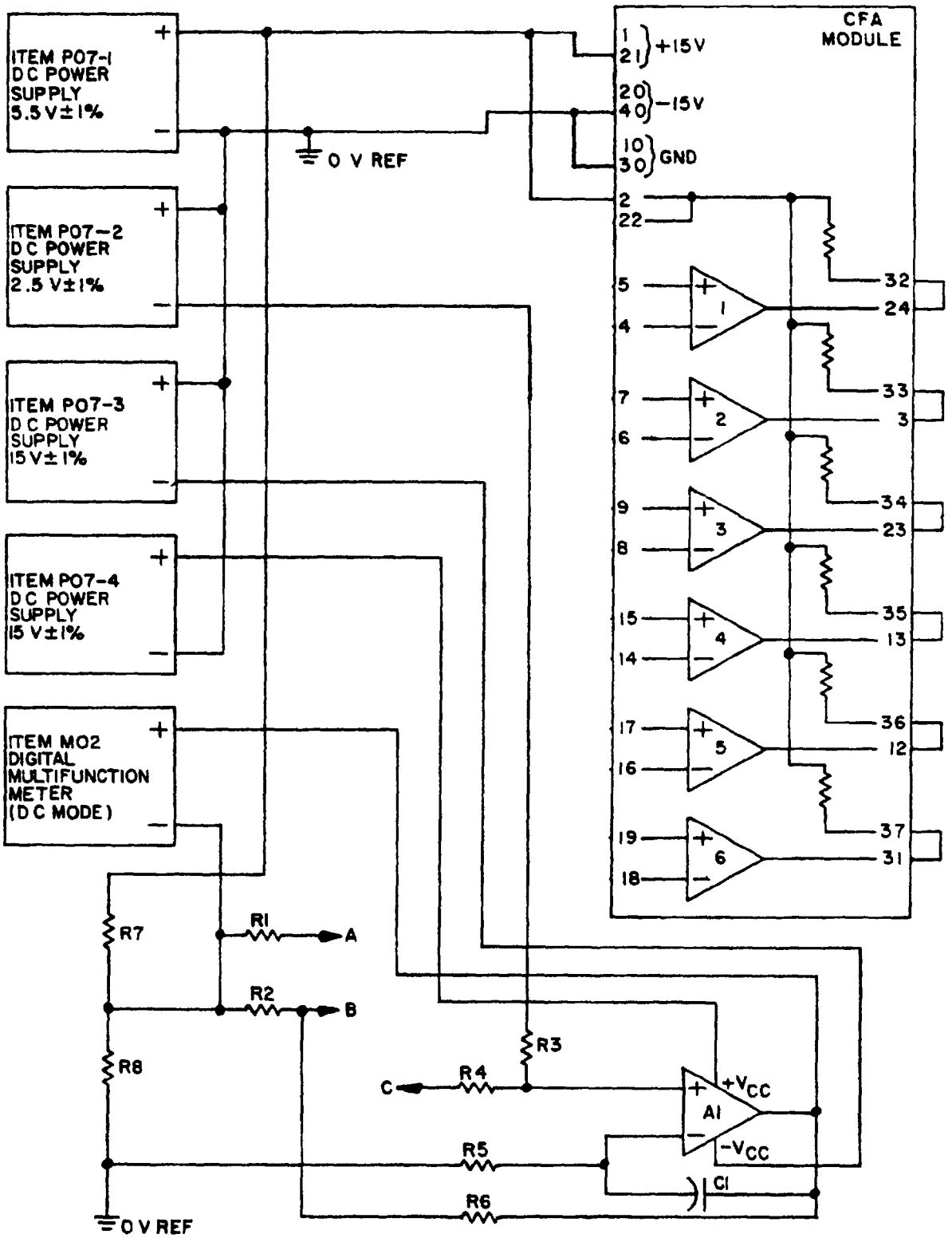


FIGURE 7. Offset voltage (+5 V) test circuit for table V (test conditions C7 through C12).

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NOTES:
1. R1 = R2 = 49.9Ω ±0.1%, 1/8 W.
2. R3 = R4 = R5 = 100 kΩ ±1%, 1/8 W.
3. R6 = 49.9 kΩ ±0.1%, 1/8 W.
4. R7 = R8 = 10 kΩ ±1%, 1/8 W.
5. C1 = 0.01 μF.
6. A1 = Operational amplifier 741, or equivalent.
7. For each test condition, divide the M02 reading by 1,000. The resultant shall be within the table V limits.

FIGURE 7. Offset voltage (+5 V) test circuit for table V (test conditions C7 through C12) - Continued.

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TABLE VI. Input bias current test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	Input bias current, ± 15 V positive terminal	5	4	24	0	100	nA	8
D2	Input bias current, ± 15 V negative terminal	5	4	24				
D3	Input bias current, ± 15 V positive terminal	7	6	3				
D4	Input bias current, ± 15 V negative terminal	7	6	3				
D5	Input bias current, ± 15 V positive terminal	9	8	23				
D6	Input bias current, ± 15 V negative terminal	9	8	23				
D7	Input bias current, ± 15 V positive terminal	15	14	13				
D8	Input bias current, ± 15 V negative terminal	15	14	13				
D9	Input bias current, ± 15 V positive terminal	17	16	12				
D10	Input bias current, ± 15 V negative terminal	17	16	12				
D11	Input bias current, ± 15 V positive terminal	19	18	31				
D12	Input bias current, ± 15 V negative terminal	19	18	31	0	100	nA	8

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TABLE VI. Input bias current test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D13	Input bias current, ± 5 V positive terminal	5	4	24	0	100	nA	9
D14	Input bias current, + 5 V negative terminal	5	4	24				
D15	Input bias current, ± 5 V positive terminal	7	6	3				
D16	Input bias current, ± 5 V negative terminal	7	6	3				
D17	Input bias current, ± 5 V positive terminal	9	8	23				
D18	Input bias current, ± 5 V negative terminal	9	8	23				
D19	Input bias current, ± 5 V positive terminal	15	14	13				
D20	Input bias current, ± 5 V negative terminal	15	14	13				
D21	Input bias current, ± 5 V positive terminal	17	16	12				
D22	Input bias current, ± 5 V negative terminal	17	16	12				
D23	Input bias current, ± 5 V positive terminal	19	18	31				
D24	Input bias current, ± 5 V negative terminal	19	18	31	0	100	nA	9

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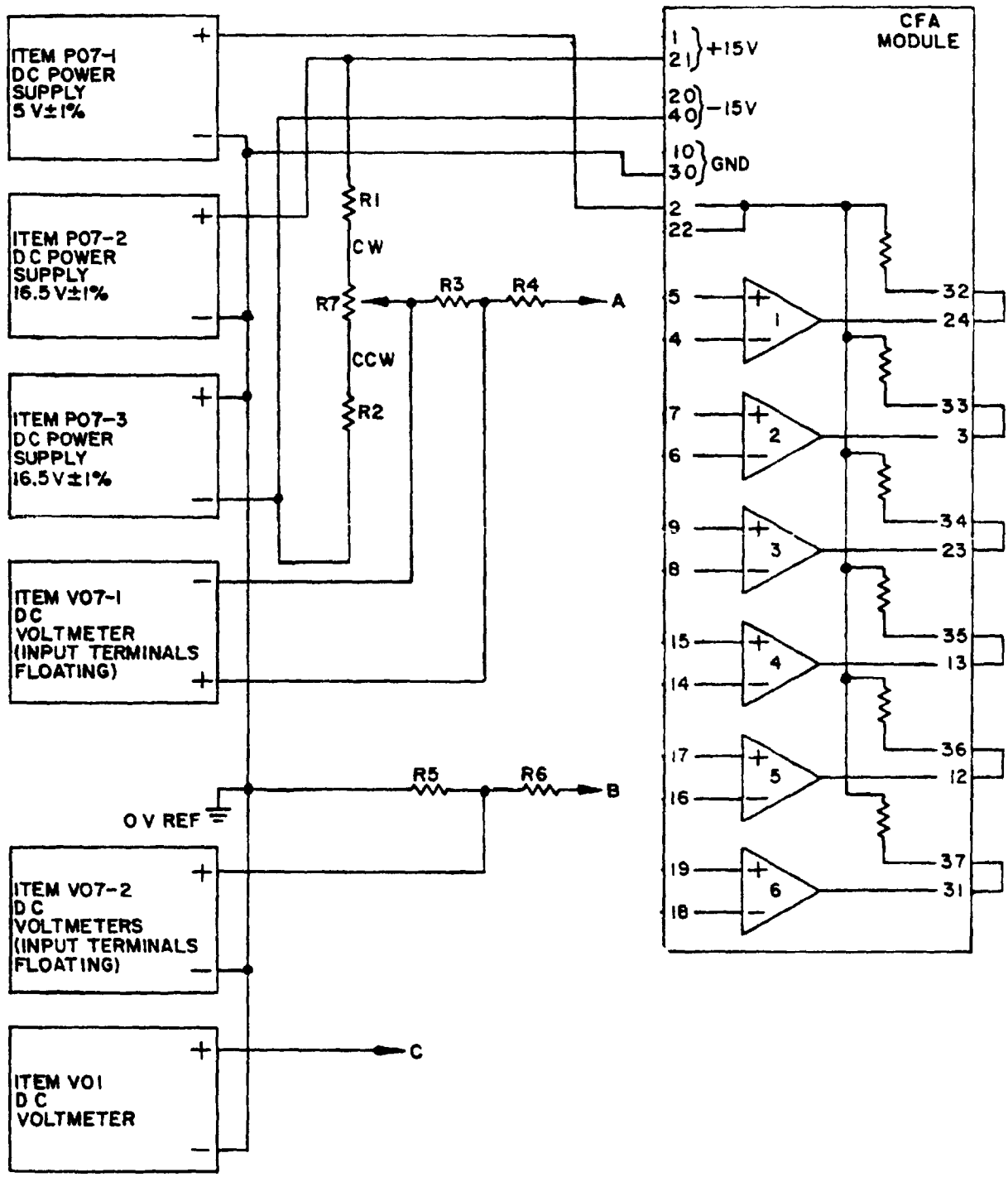


FIGURE 8. Input bias current (±15 V) test circuit for table VI (test conditions D1 through D12).

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NOTES
1 R1-R6 = 10.0 kΩ ±1%, 1/8 W.
2. R7 = 500Ω, 0.5 W potentiometer.
3 Before making measurement on each circuit (before test conditions D1, D3, D5, etc.) adjust R7 as follows

(a) Adjust R7 CCW until the comparator output, as measured with item V01, is between 0 and 1.0 Vdc.
(b) Slowly adjust R7 CW until the comparator just goes to between 4 5 and 5 2 Vdc

4. Make a voltage measurement with item V07-1 for test conditions D1 through D11 (odd numbers) and make a voltage measurement with item V07-2 for test conditions D2 through D12 (even numbers) The bias current shall be determined as follows

$$\text{Bias current} = \frac{V}{10}$$

where V is the voltage in μV
10 is the resistance in kΩ
and Bias current is in nA.

The resultant shall be within the limits specified.
5. Since the item V07 voltmeters are not used simultaneously only one is required.

FIGURE 8 Input bias current (±15 V) test circuit for table VI (test conditions D1 through D12) - Continued

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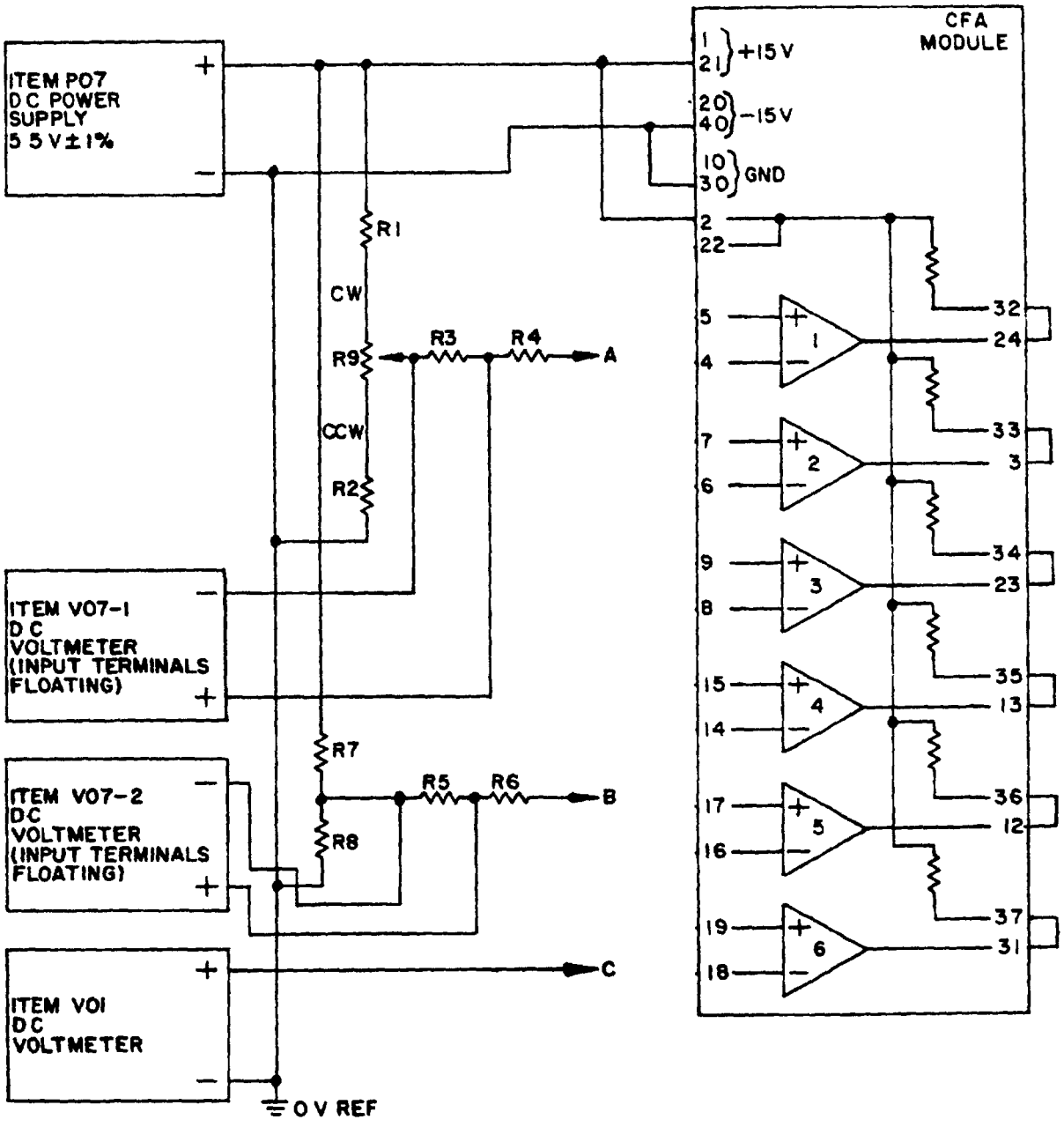


FIGURE 9. Input bias current (+5 V) test circuit for table VI (test conditions D13 through D24).

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NOTES
1. R1-R8 = 10.0 kΩ ±1%, 1/8 W.
2. R9 = 500Ω, 0.5 W potentiometer.
3. Before making measurements on each circuit (before test conditions D13, D15, D17, etc.) adjust R9 as follows.

   (a) Adjust R9 CCW until the comparator output, as measured with item V01, is between 0 and 1.0 Vdc.
   (b) Slowly adjust R9 CW until the comparator just goes to between 4.5 and 5.2 Vdc.

4. Make a voltage measurement with item V07-1 for test conditions D13 through D23 (odd numbers) and make a voltage measurement with item V07-2 for test conditions D14 through D24 (even numbers). The bias current shall be determined as follows

$$\text{Bias current} = \frac{V}{10}$$

where V is the voltage in μV
10 is the resistance in kΩ
and Bias current is in nA.

   The resultant shall be within the limits specified.
5 Since the item V07 voltmeters are not used simultaneously only one is required

FIGURE 9 Input bias current (+5 V) test circuit for table VI (test conditions D13 through D24) - Continued

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TABLE VII. Input offset current check

Test condition	Parameter	Calculations			Initial limits 25° ±5°C Ambient		Unit	Test figure
		Test condition	minus	Test condition	Min	Max		
E1	Input offset current (± 15 V)	D1		D2	- 10	+ 10	nA	- -
E2	↓	D3		D4	↓	↓	↓	↓
E3	↓	D5		D6	↓	↓	↓	↓
E4	↓	D7		D8	↓	↓	↓	↓
E5	↓	D9		D10	↓	↓	↓	↓
E6	Input offset current (± 15 V)	D11		D12	↓	↓	↓	↓
E7	Input offset current (+ 5 V)	D13		D14	↓	↓	↓	↓
E8	↓	D15		D16	↓	↓	↓	↓
E9	↓	D17		D18	↓	↓	↓	↓
E10	↓	D19		D20	↓	↓	↓	↓
E11	↓	D21		D22	↓	↓	↓	↓
F12	Input offset current (+ 5 V)	D23		D24	- 10	+ 10	nA	- -

NOTE: No measurements are taken for this check. The results of test conditions D1 through D24 are used in the calculations indicated above. The resultants of the calculations shall meet the specified limits

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TABLE VIII. Low-level output voltage test

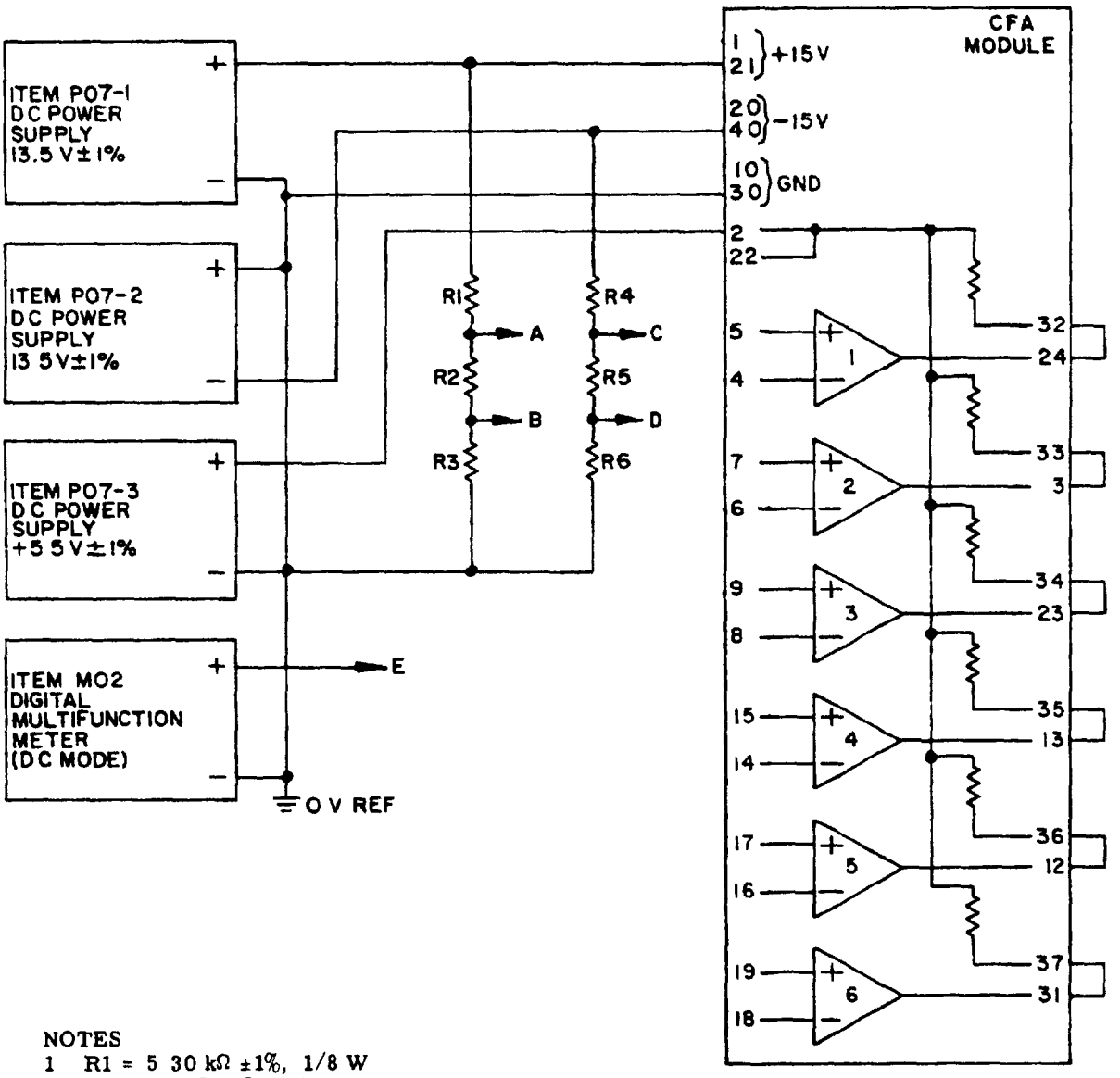
Test condition	Parameter	Connect					Initial limits 25° +5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
F1	Low-level output voltage I_o = 5.5 mA, V_{IN} = 12.2 Vdc	4	5	Open	Open	24	0.01	0.40	V	10
F2	↓	6	7	↓	↓	3	↓	↓	↓	↓
F3	↓	8	9	↓	↓	23	↓	↓	↓	↓
F4	↓	14	15	↓	↓	13	↓	↓	↓	↓
F5	↓	16	17	↓	↓	12	↓	↓	↓	↓
F6	Low-level output voltage I_o = 5.5 mA, V_{IN} = 12.2 Vdc	18	19	Open	Open	31	↓	↓	↓	↓
F7	Low level output voltage I_o = 5.5 mA, V_{IN} = -13.1 Vdc	Open	Open	5	4	24	↓	↓	↓	↓
F8	↓	↓	↓	7	6	3	↓	↓	↓	↓
F9	↓	↓	↓	9	8	23	↓	↓	↓	↓
F10	↓	↓	↓	15	14	13	↓	↓	↓	↓
F11	↓	↓	↓	17	16	12	↓	↓	↓	↓
F12	Low level output voltage I_o = 5.5 mA, V_{IN} = -13.1 Vdc	Open	Open	19	18	31	0.01	0.40	V	10

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TABLE VIII. Low-level output voltage test - Continued

Test condition	Parameter	Connect					Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
F13	Low-level output voltage I_o = 50 mA, V_{IN} = -13.1 Vdc	Open	Open	5	4	24	0 1	1.50	V	11
F14				7	6	3				
F15				9	8	23				
F16				15	14	13				
F17				17	16	12				
F18	Low-level output voltage I_o = 50 mA, V_{IN} = -13.1 Vdc	Open	Open	19	18	31	0 1	1.50		11
F19	Low-level output voltage I_o = 8 mA, V_{IN} = 0 4 Vdc	4	5	Open	Open	24	0.01	0.40		12
F20		6	7			3				
F21		8	9			23				
F22		14	15			13				
F23		16	17			12				
F24	Low-level output voltage I_o = 8 mA, V_{IN} = 0.4 Vdc	18	19	Open	Open	31	0.01	0 40	V	12

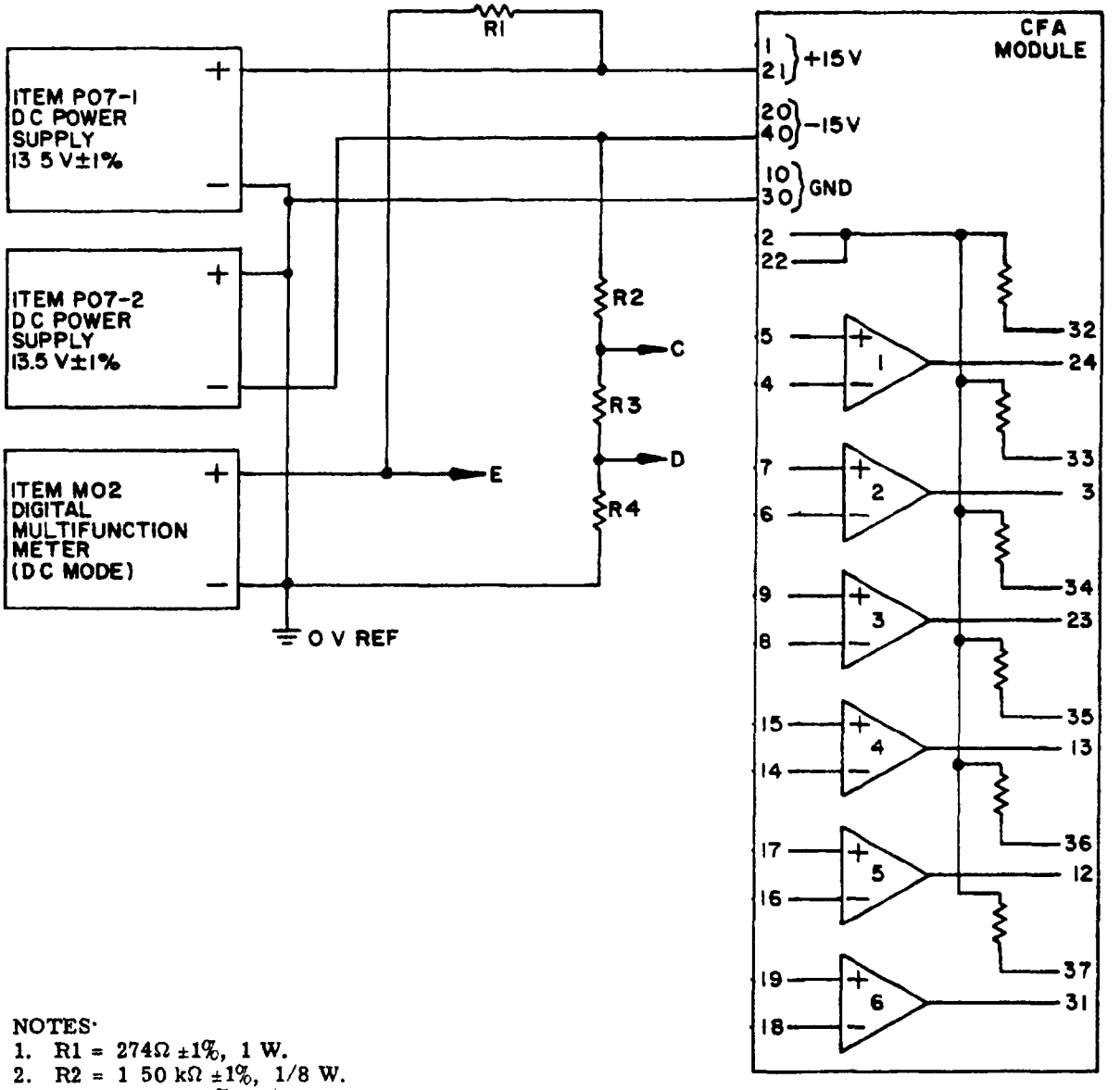
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NOTES
1. R1 = 5 30 kΩ ±1%, 1/8 W
2. R2 = 25 5Ω ±1%, 1/8 W.
3. R3 = R6 = 49 9 kΩ ±1%, 1/8 W
4. R4 = 1 50 kΩ ±1%, 1/8 W
5. R5 = 23 7Ω ±1%, 1/8 W.
6. The voltage at points A and B shall be between 12 05 V and 12 35 V, however, the difference between points A and B shall be 6 2 ±0 2 mV.
7. The voltage at points C and D shall be between -13 24 V and -12.97 V, however, the difference between points C and D shall be 6 2 ±0 2 mV.
8. Test conditions F1-F12 verify the low-level output voltage of 0.40 V maximum and also the common mode voltage range of between ($-V_{CC}$ + 0.4 V) and ($+V_{CC}$ − 1.3 V).

FIGURE 10. Low-level output voltage test circuit for table VIII (test conditions F1 through F12).

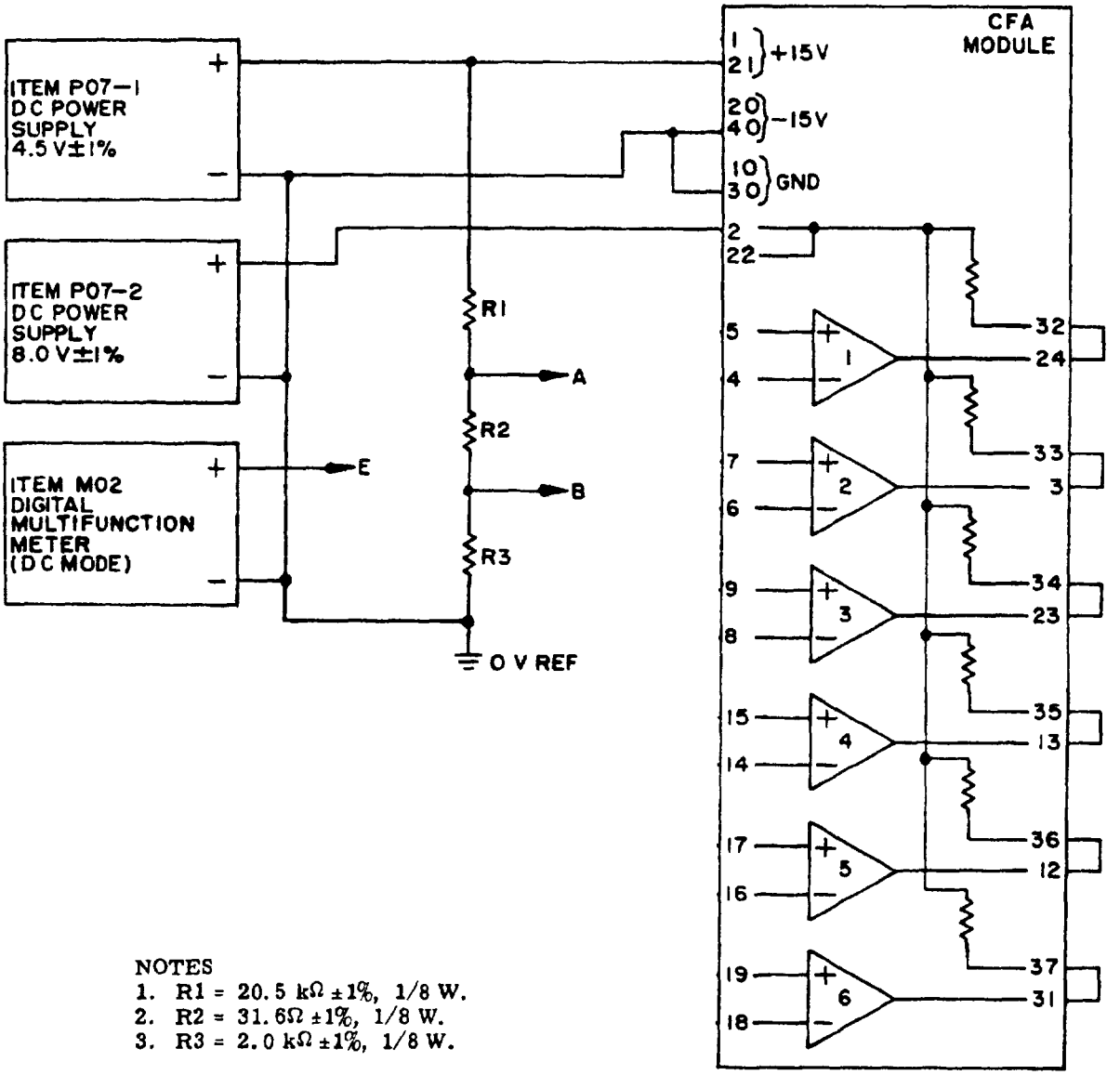
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NOTES:
1. R1 = 274Ω ±1%, 1 W.
2. R2 = 1 50 kΩ ±1%, 1/8 W.
3. R3 = 23. 7Ω ±1%, 1/8 W.
4. R4 = 49. 9 kΩ ±1%, 1/8 W.

FIGURE 11. Low-level output voltage test circuit for table VIII (test conditions F13 through F18).

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NOTES
1. R1 = 20.5 kΩ ±1%, 1/8 W.
2. R2 = 31.6Ω ±1%, 1/8 W.
3. R3 = 2.0 kΩ ±1%, 1/8 W.

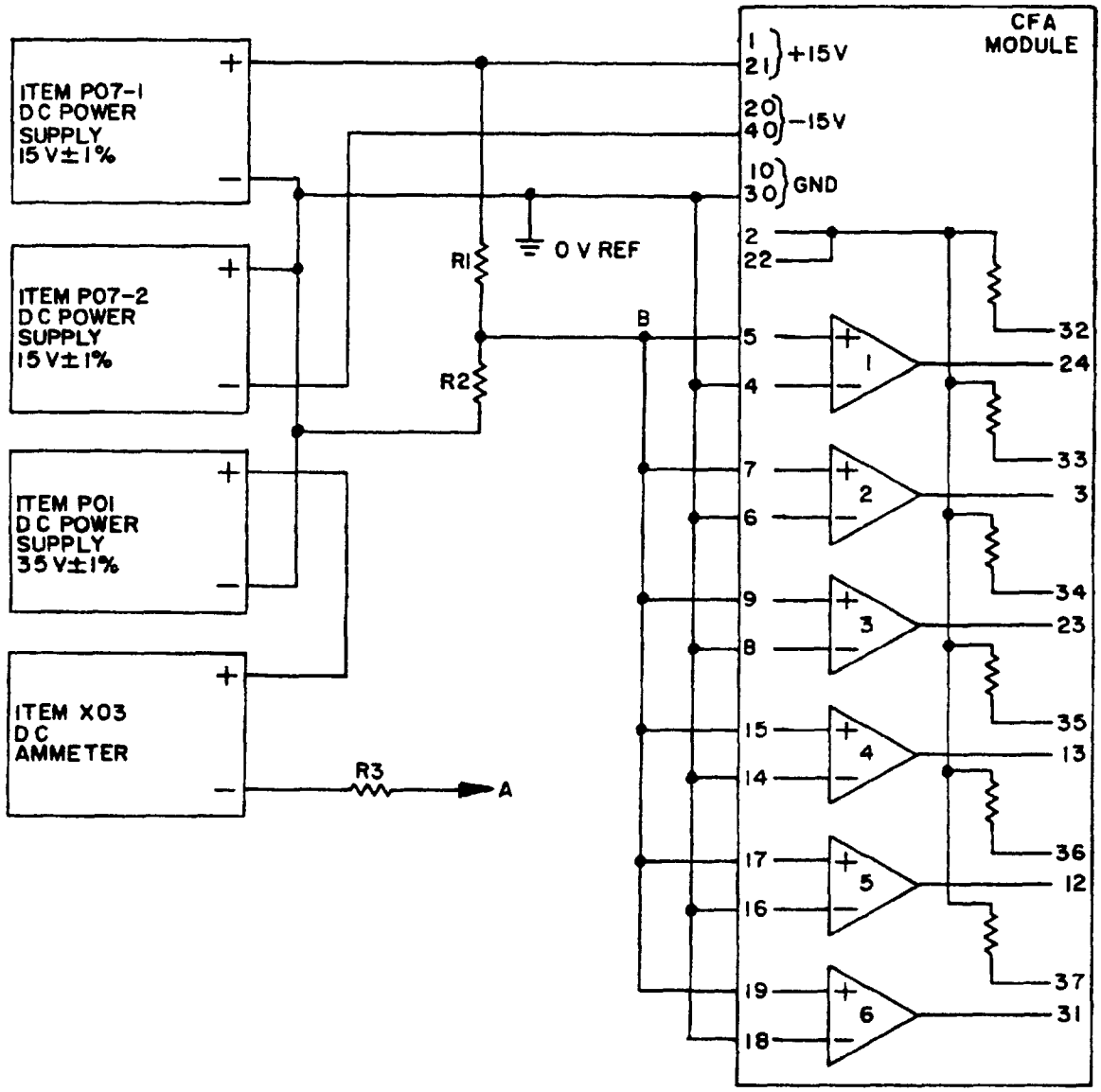
FIGURE 12. Low-level output voltage test circuit for table VIII (test conditions F19 through F24).

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TABLE IX. Output leakage current test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
G1	Output leakage current	24	0	50	nA	13
G2	↓	3	↓	↓	↓	↓
G3		23				
G4		13				
G5		12				
G6	Output leakage current	31	0	50	nA	13

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NOTES:
1. R1 = 249 kΩ ±1%, 1/8 W.
2. R2 = 100Ω ±1%, 1/8 W.
3. R3 = 5.36 kΩ ±1%, 1/8 W.
4. The voltage at point B shall be 6.0 ±0.2 mV.

FIGURE 13. Output leakage current test circuit for table IX.

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TABLE X. Resistance and continuity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Resistance (R1)	2	32	925	1075	Ω	14
H2	Resistance (R2)	↓	33	↓	↓	↓	↓
H3	Resistance (R3)	↓	34	↓	↓	↓	↓
H4	Resistance (R4)	↓	35	↓	↓	↓	↓
H5	Resistance (R5)	↓	36	↓	↓	↓	↓
H6	Resistance (R6)	2	37	925	1075	↓	↓
J1	Continuity	1	21	0	1.0	↓	↓
J2	↓	20	40	↓	↓	↓	↓
J3	↓	10	30	↓	↓	↓	↓
J4	Continuity	2	22	0	1.0	Ω	14

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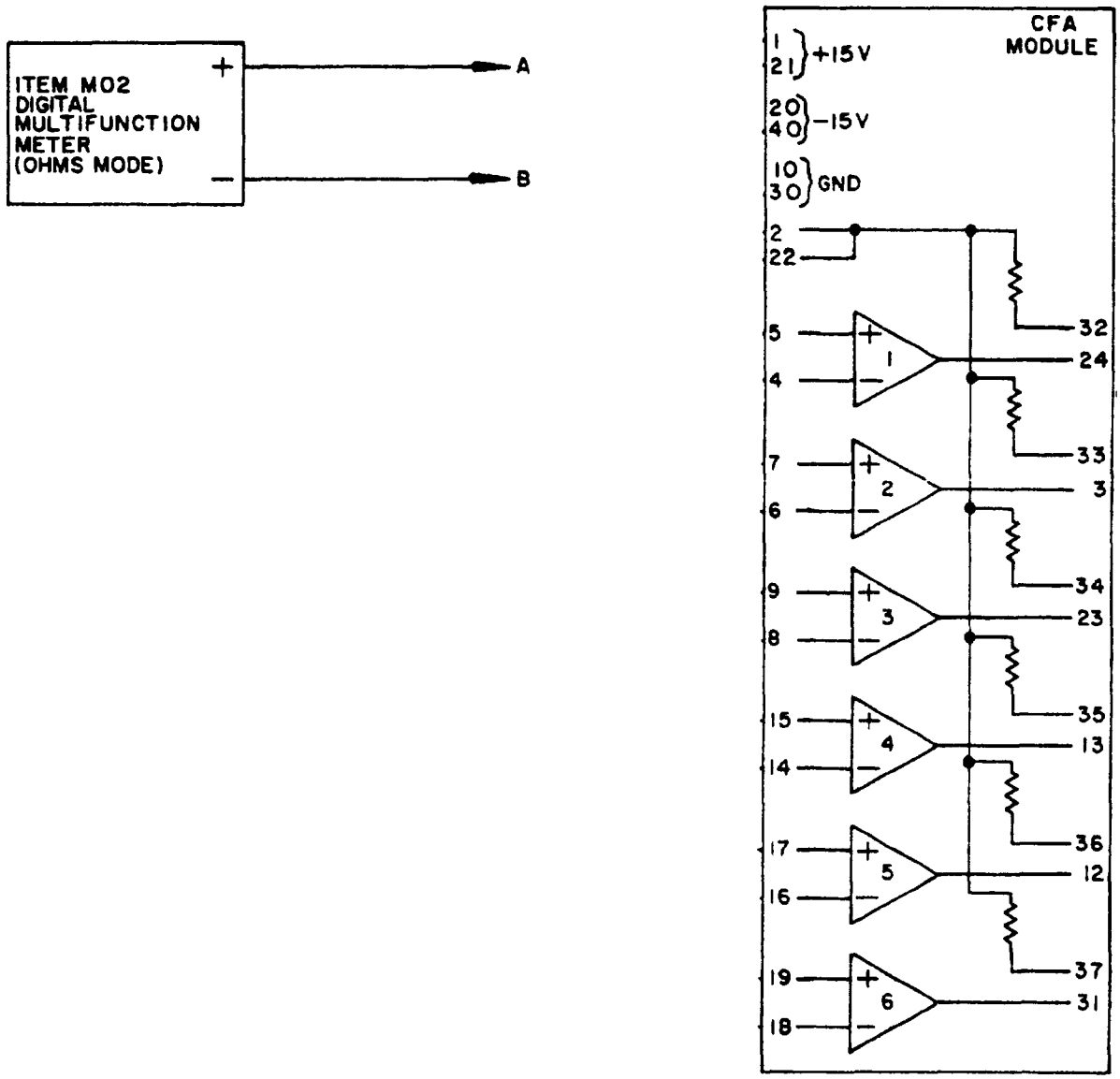


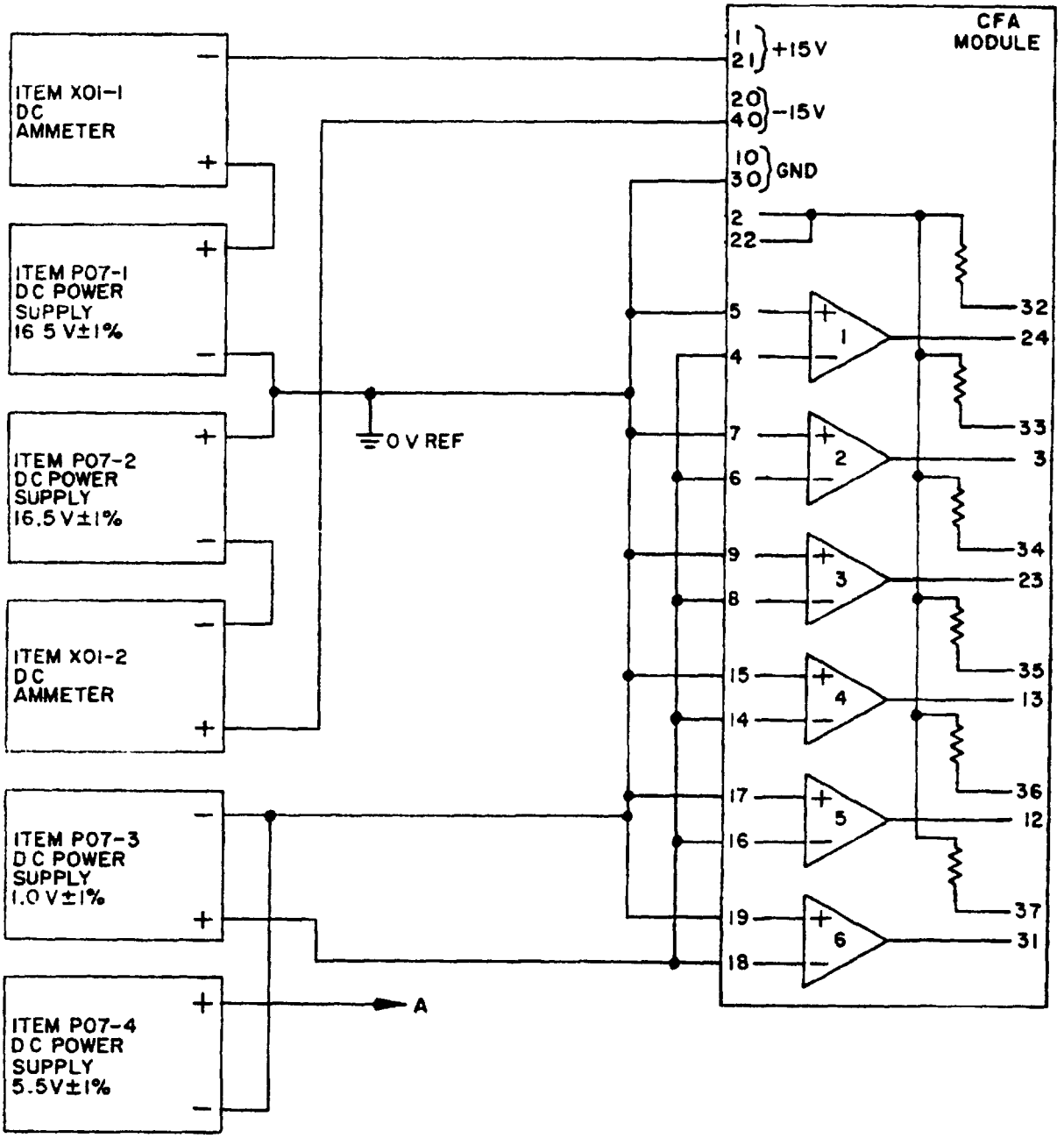
FIGURE 14 Resistance and continuity test circuit for table X.

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TABLE XI Power supply current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Pin to pin	Min	Max		
K1	Power supply current (+ 15 V), pullup resistors not connected	Open		1.0	38.0	mA	15
K2	Power supply current (- 15 V), pullup resistors not connected	Open			31 5		
K3	Power supply current (+ 15 V), pullup resistors connected	2	32 to 24, 33 to 3, 34 to 23, 35 to 13, 36 to 12, 37 to 31		38 0		
K4	Power supply current (- 15 V), pullup resistors connected	2	32 to 24, 33 to 3, 34 to 23, 35 to 13, 36 to 12, 37 to 31		31.5		15
K5	Power supply current (+ 5 V)	Open		1.0	31.5	mA	16

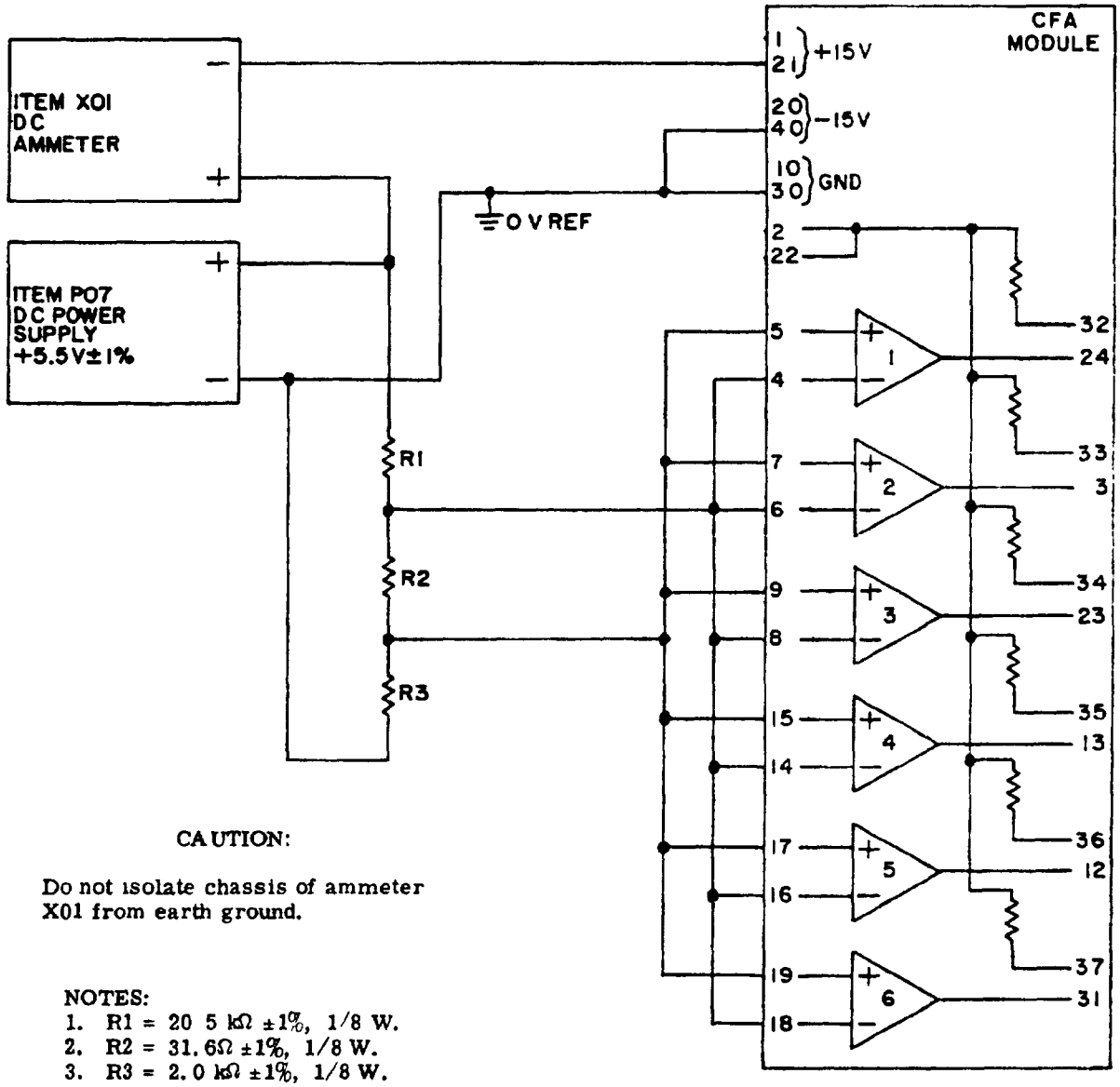
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CAUTION Do not isolate chassis of ammeter X01 from earth ground.

FIGURE 15 Power supply current (+15 V) test circuit for table XI (test conditions K1 through K4).

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CAUTION:

Do not isolate chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 20 5 kΩ ±1%, 1/8 W.
2. R2 = 31.6Ω ±1%, 1/8 W.
3. R3 = 2.0 kΩ ±1%, 1/8 W.

FIGURE 16. Power supply current (+5 V) test circuit for table XI (test condition K5).

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TABLE III Response time (t_{RLH} and t_{RHL}) test.

Test condition	Parameter	Connect			Initial limits 25° +5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	Response time, t_{RLH}	4	5	24	30	400	ns	17, 18
L2	↓	6	7	3	↓	↓	↓	↓
L3		8	9	23				
L4		14	15	13				
L5	↓	16	17	12				↓
L6	Response time, t_{RLH}	18	19	31				17, 18
L7	Response time, t_{RHL}	4	5	24				19, 20
L8	↓	6	7	3				↓
L9		8	9	23				
L10		14	15	13				
L11	↓	16	17	12	↓	↓	↓	↓
L12	Response time, t_{RHL}	18	19	31	30	400	ns	19, 20

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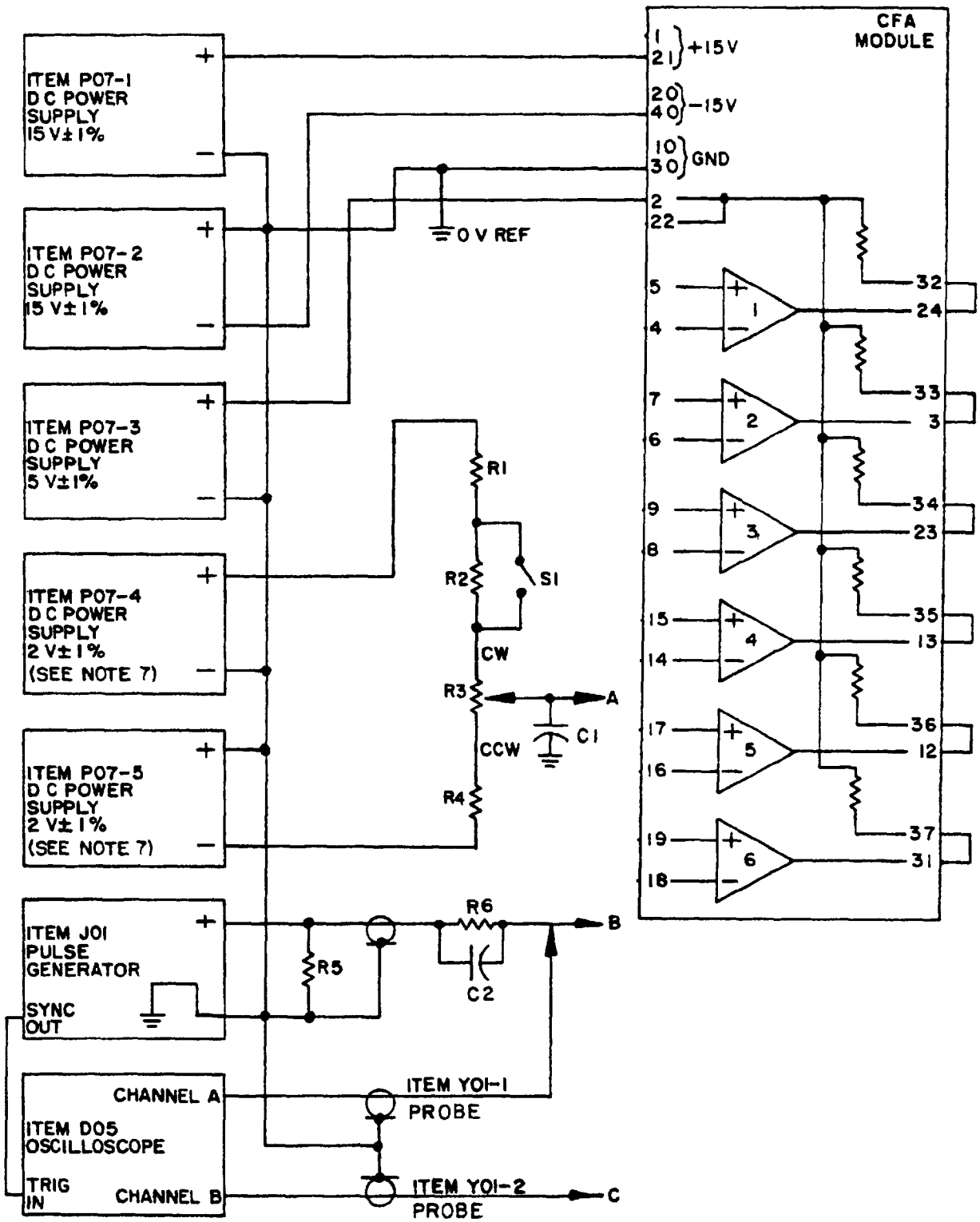


FIGURE 17 Response time (t_{RLH}) test circuit for table XII (test conditions L1 through L6).

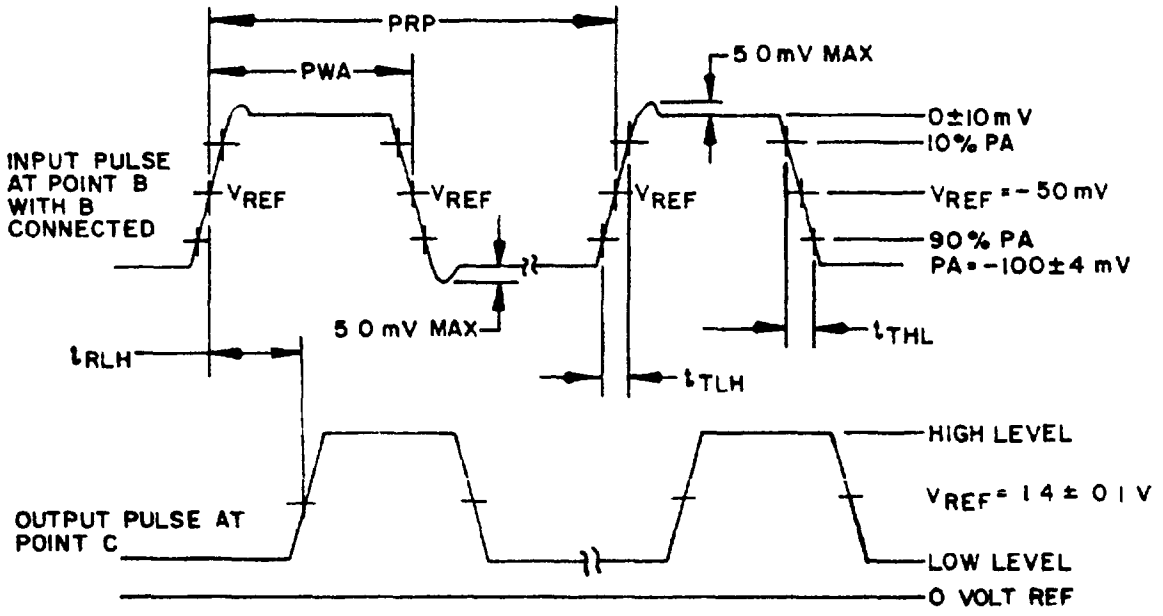
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NOTES
1. R1 = R4 = 5.36 kΩ ±1%, 1/8 W
2. R2 = 27.4Ω ±1%, 1/8 W.
3. R3 = 100Ω, 3/4 W potentiometer
4. R5 = 51Ω ±5%, 1/2 W.
5. R6 = 2.7 kΩ ±1%, 1/8 W.
6. C1 = C2 = 0.001 µF ±10%
7. Use item D05 to set items P07-4 and P07-5 to 2 V ±1%
8. For each test condition L1 through L6, conduct the test as follows

   (a) With all points connected close S1
   (b) Adjust R3 CCW until a pulse appears at point C
   (c) Adjust R3 CW until point C just becomes a constant 0.3 ±0.3 Vdc
   (d) Open switch S1 and measure response time t_{RLH}

FIGURE 17. Response time (t_{RLH}) test circuit for table XII (test conditions L1 through L6) - Continued.

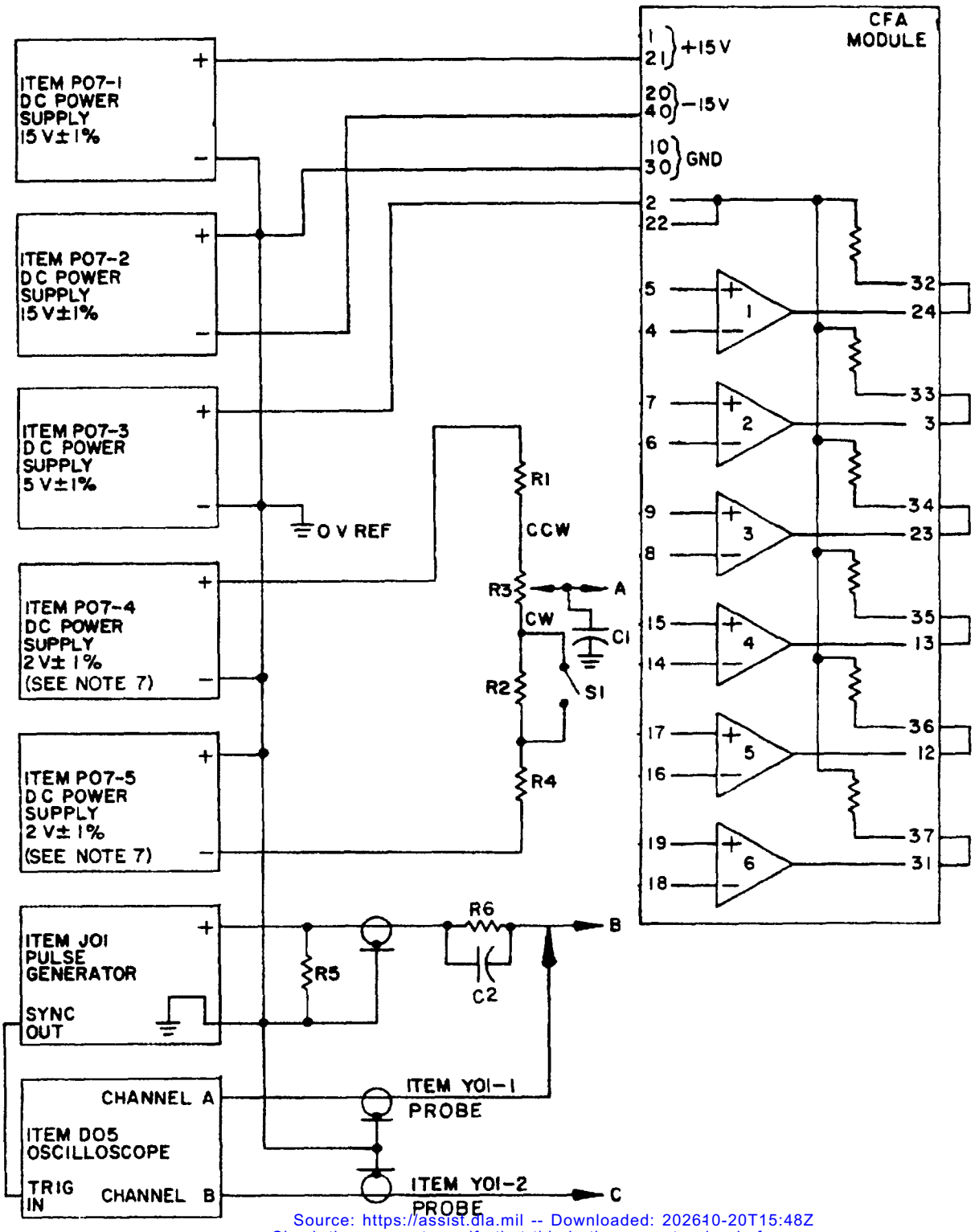
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NOTES
1. Input pulse parameters
   $t_{TLH} = t_{THL} = 5 \pm 1$ ns.
   PWA = 500 ±10 ns
   PRP = 10 ±1 µs.
2. Before making measurements specified in table XII, verify that the output pulse high level is ≥ 4.5 V and the low level is ≤ 0.6 V.
3. All voltages are referenced to ground

FIGURE 18. Response time (t_{RLH}) test circuit waveforms for table XII (test conditions L1 through L6).

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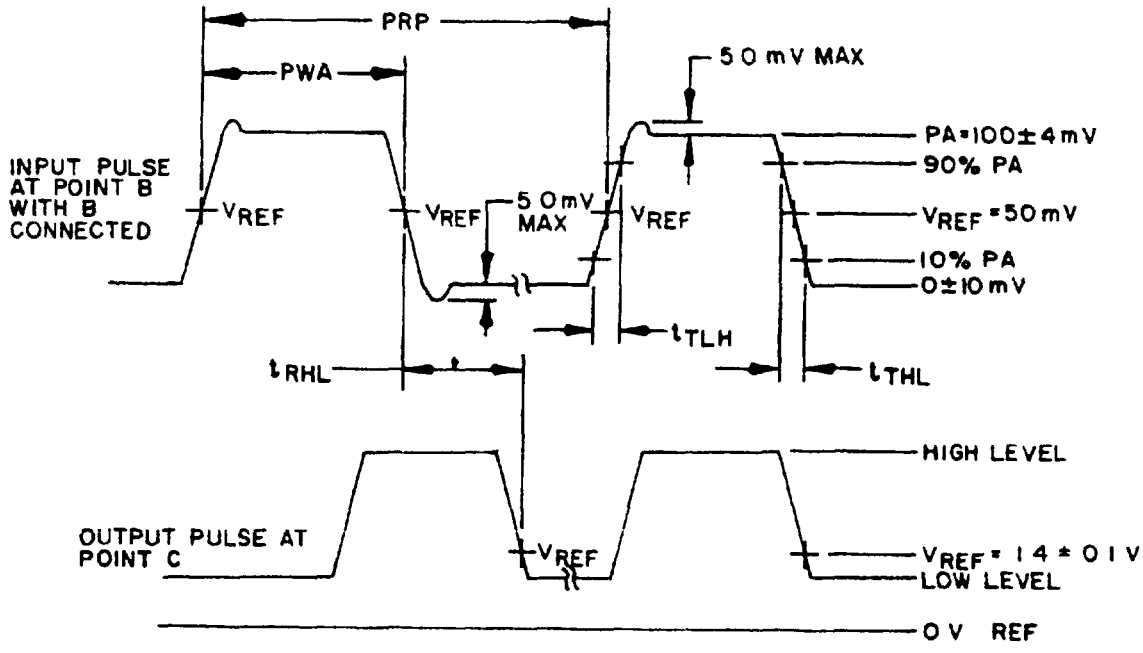
FIGURE 19. Response time (t_{RHL}) test circuit for table XII (test conditions L7 through L12).

NOTES
1. R1 = R4 = 5.36 kΩ ±1%, 1/8 W.
2. R2 = 27.4Ω ±1%, 1/8 W.
3 R3 = 100Ω, 3/4 W potentiometer.
4 R5 = 51Ω ±5%, 1/2 W.
5 R6 = 2 7 kΩ ±1%, 1/8 W.
6 C1 = C2 = 0 001 μF ±10%.
7. Use item D05 to set items P07-4 and P07-5 to 2 V ±1%
8 For each test condition L7 through L12, conduct the test as follows:

(a) With all points connected close S1.
(b) Adjust R3 CCW until a pulse appears at point C.
(c) Adjust R3 CW until point C just becomes a constant 5 0 ±0 5 Vdc.
(d) Open S1 and measure response time t_{RHL}

FIGURE 19. Response time (t_{RHL}) test circuit for table XII (test conditions L7 through L12) - Continued.

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NOTES
1 Input pulse parameters:
$t_{TLH} = t_{THL} = 5 \pm 1$ ns.
PWA = 500 ±10 ns
PRP = 10 ±1 µs.
2 Before making measurements specified in table XII, verify that the output pulse high level is ≥ 4 5 V and the low level is ≤ 0 6 V.
3 All voltages are referenced to ground.

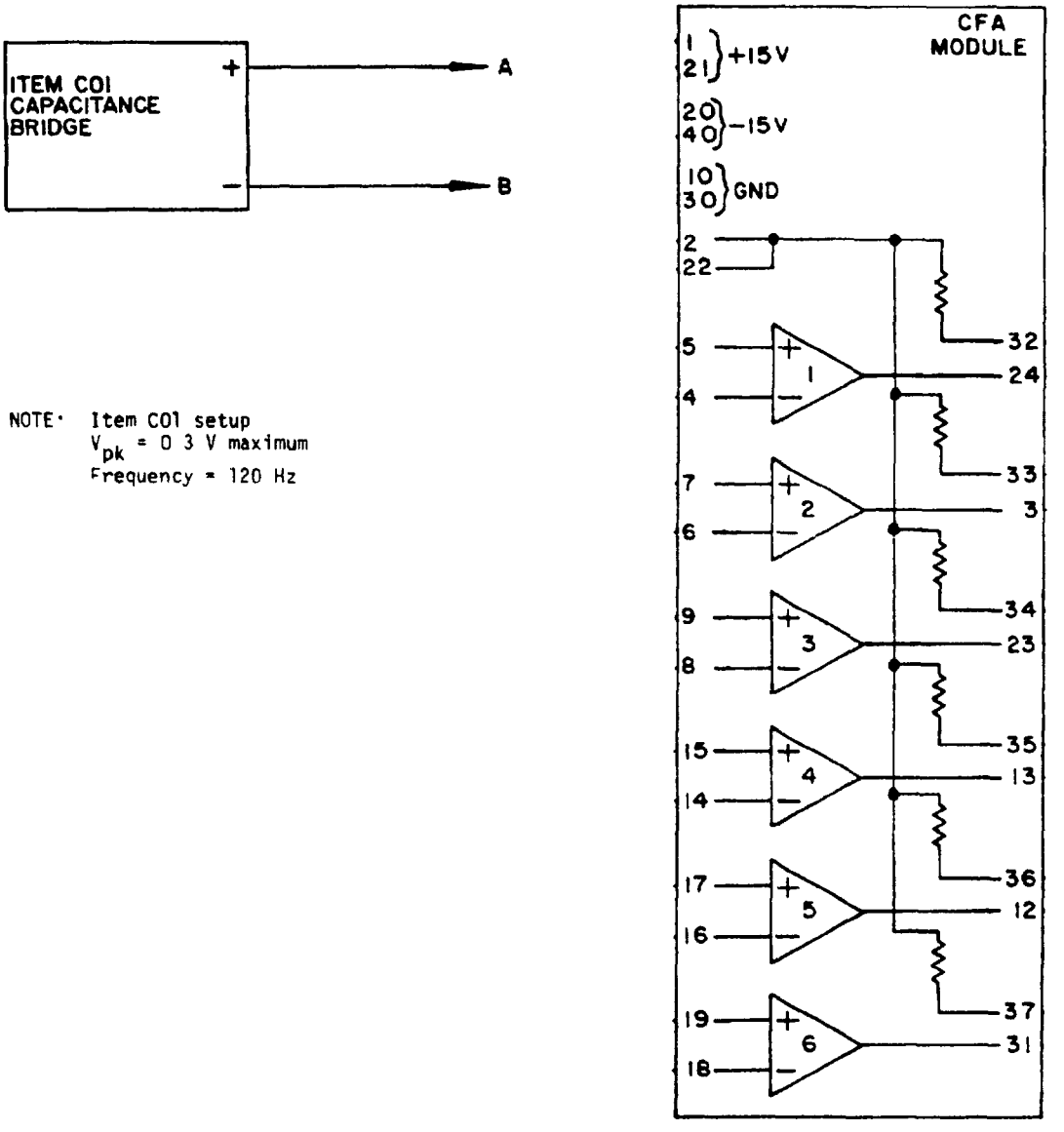
FIGURE 20. Response time (t_{RHL}) test circuit waveforms for table XII (test conditions L7 through L12)

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TABLE XIII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25° +5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Power supply filter capacitance (+ 15 V)	1	10	0.75	1.25	µF	21
M2	Power supply filter capacitance (- 15 V)	20	10	0.75	1.25	↓	21
M3	Power supply filter capacitance (pin 2)	2	10	0.006	0.012	µF	21

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NOTE: Item C01 setup
V_{pk} = 0 3 V maximum
Frequency = 120 Hz

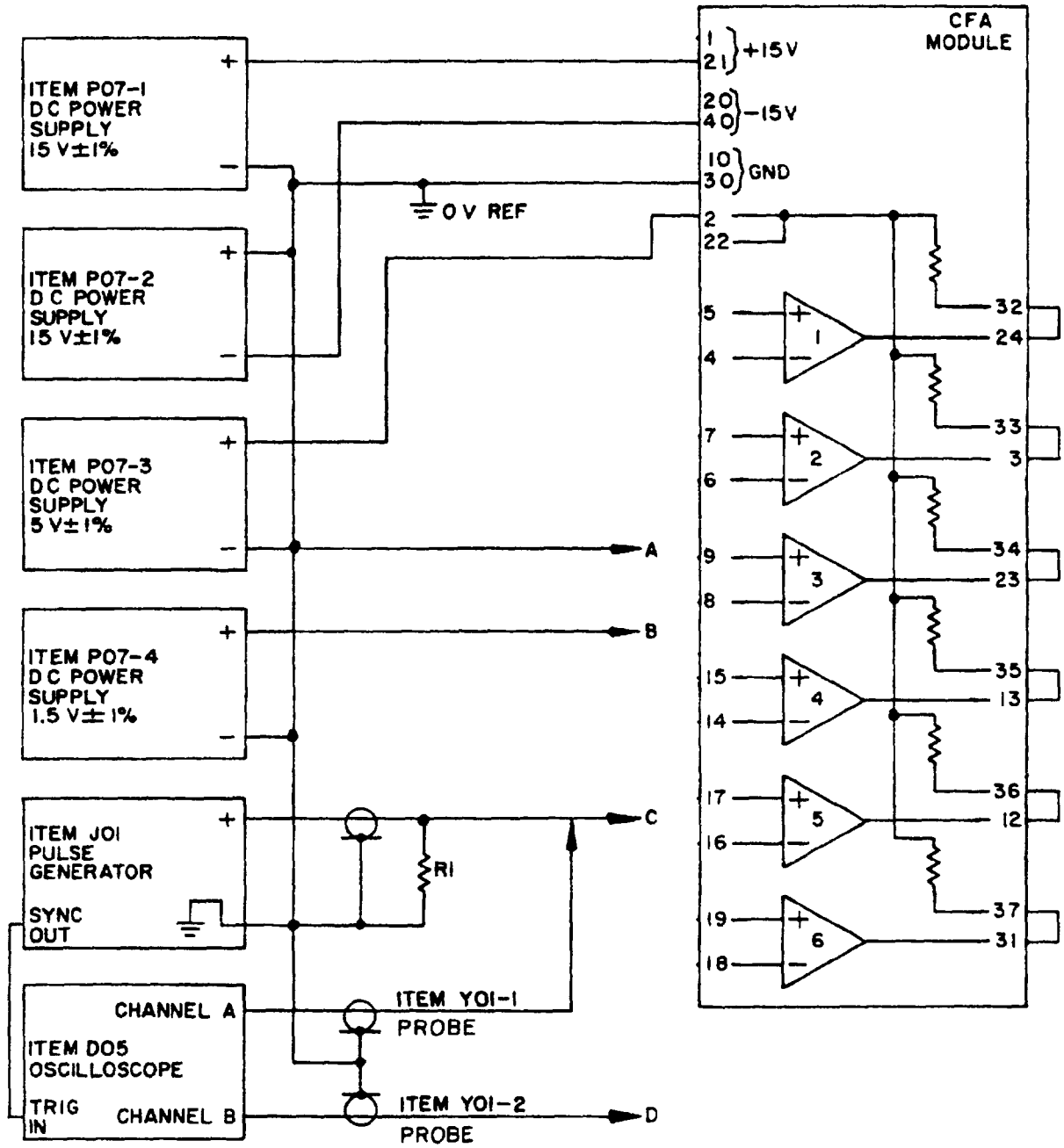
FIGURE 21 Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	Output crosstalk, low level	5	4,6,8, 14,16,18	7,9,15, 17,19	24	0	500	mV	22, 23
N2		7		5,9,15, 17,19	3				
N3		9		5,7,15, 17,19	23				
N4		15		5,7,9, 17,19	13				
N5		17		5,7,9, 15,19	12				
N6	Output crosstalk, low level	19	4,6,8, 14,16,18	5,7,9, 15,17	31		500		
N7	Output crosstalk, high level	4	5,7,9, 15,17,19	6,8,14, 16,18	24		400		
N8		6		4,8,14, 16,18	3				
N9		8		4,6,14, 16,18	23				
N10		14		4,6,8, 16,18	13				
N11		16		4,6,8, 14,18	12				
N12	Output crosstalk, high level	18	5,7,9, 15,17,19	4,6,8, 14,16	31	0	400	mV	22, 23

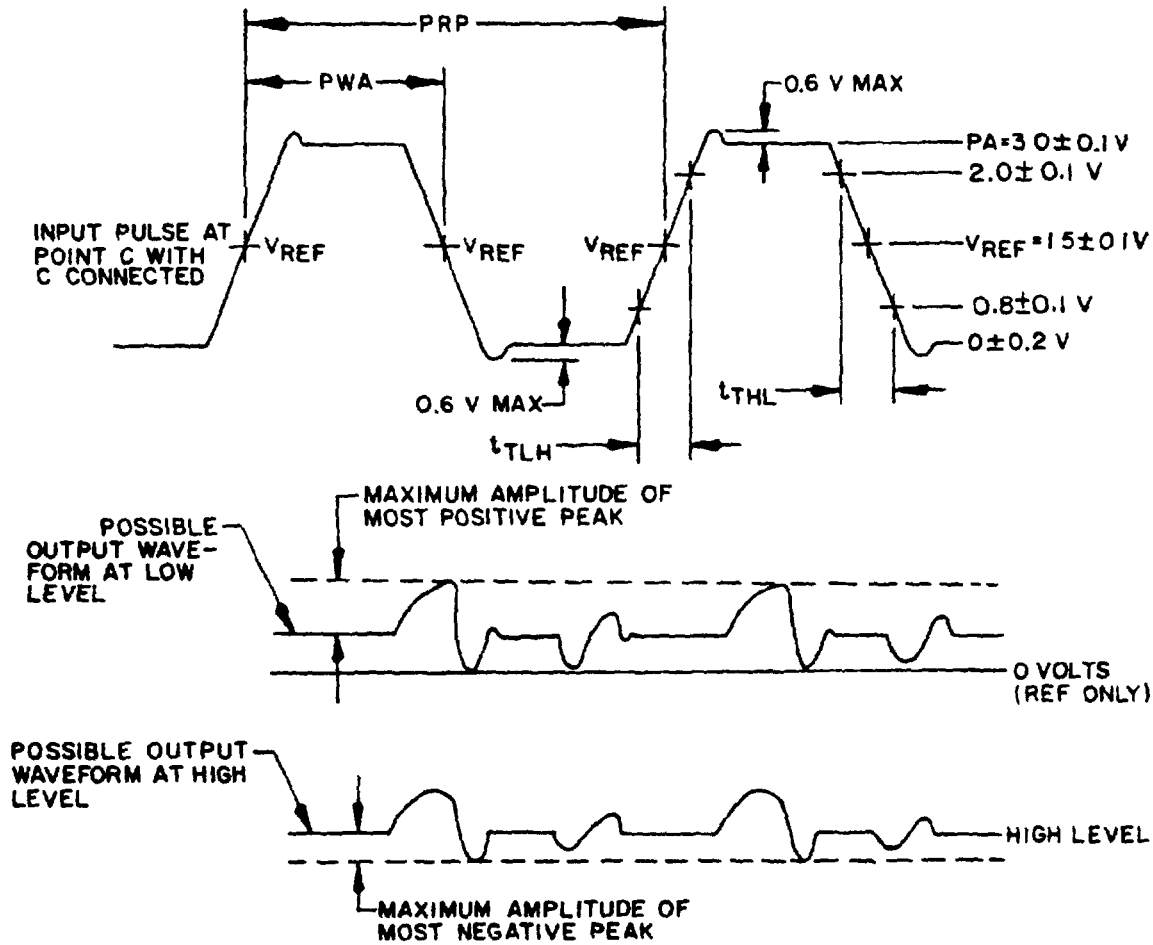
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NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 22. Output crosstalk (low level and high level) test circuit for table XIV.

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NOTES:

1. Input pulse parameters:
   For each test condition with the module connected to the pulse generator, adjust the pulse generator for the required pulse amplitude, pulse low level voltage, and transition times. Verify the input pulse parameters at each input pin before making measurement.

   $t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns.
   PWA = 5 ±0.5 µs.
   PRP = 10 ±1 µs

2. The lead length of electrical conductors shall be less than 1 inch (25.4 mm) between terminator resistor R1 and input pins, probes Y01-1 and Y01-2 to input and output pins, and ground leads of probes Y01-1 and Y01-2 to test fixture and module ground pins.

FIGURE 23. Output crosstalk (low-level and high-level) test circuit waveforms for table XIV.

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TABLE XV. Response time (t_{RLH} and t_{RHL}), V_{CC} = 30 V, test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	Response time (t_{RLH}), V_{CC} = 30 V	4	5	24	30	400	ns	24,25
P2		6	7	3				
P3		8	9	23				
P4		14	15	13				
P5		16	17	12				
P6	Response time (t_{RLH}), V_{CC} = 30 V	18	19	31				24,25
P7	Response time (t_{RHL}), V_{CC} = 30 V	4	5	24				24,26
P8		6	7	3				
P9		8	9	23				
P10		14	15	13				
P11		16	17	12				
P12	Response time (t_{RHL}), V_{CC} = 30 V	18	19	31	30	400	ns	24,26

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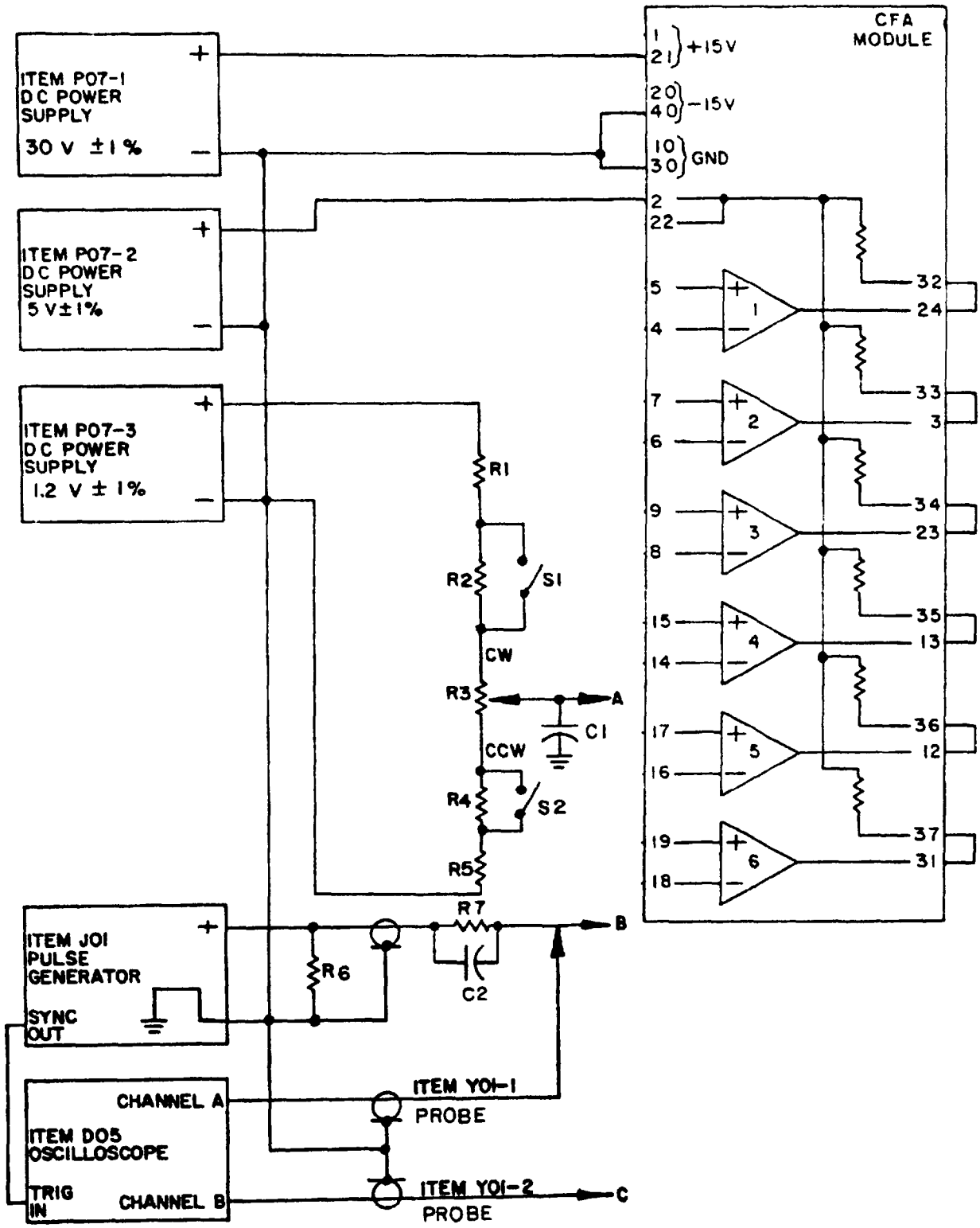


FIGURE 24 Response time (t_{RLH} and t_{RHL}) test circuit for table XV

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NOTES:
1. R1 = R5 = 2.49 kΩ ±1%, 1/8 W
2. R2 = R4 = 49.9Ω ±1%, 1/8 W
3. R3 = 100Ω, 3/4 W potentiometer.
4. R6 = 50Ω ±5%, 1 W
5. R7 = 1.3 kΩ ±1%, 1/8 W.
6. C1 = C2 = 0.001 µF ±10%.
7. For each test condition P1 through P6, conduct the test as follows:

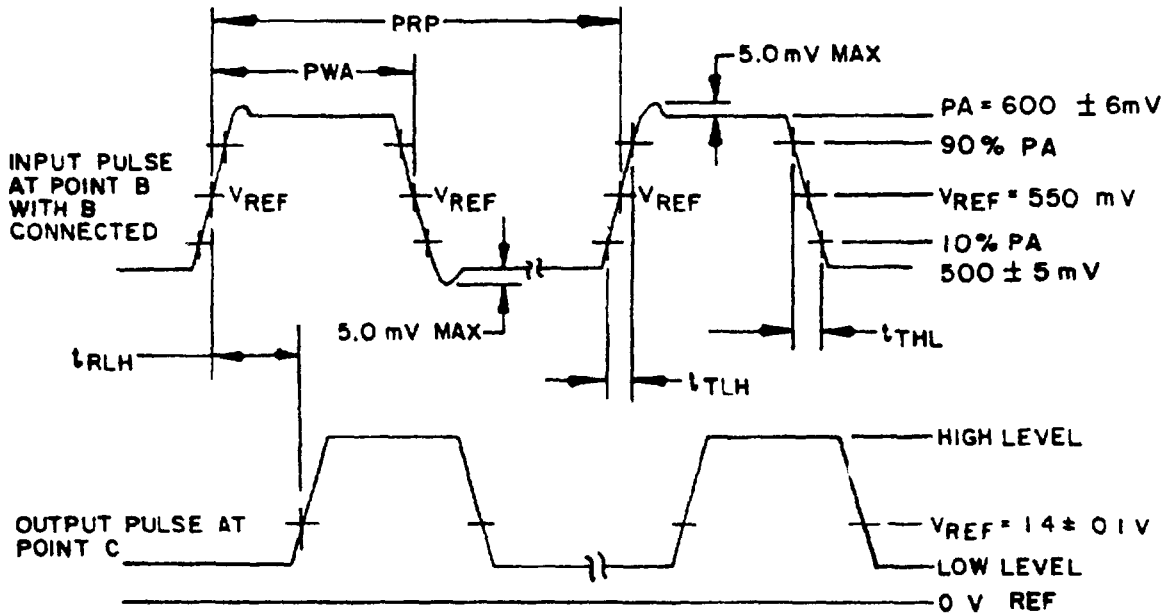
   a. With all points connected close S1 and S2.
   b. Adjust R3 CCW until a pulse appears at point C
   c. Adjust R3 CW until point C just becomes a constant 0.3 ±0.3 Vdc.
   d. Open switch S1 and measure response time t_{RLH}

8. For each test condition P7 through P12, conduct the test as follows

   a. With all points connected close S1 and S2.
   b. Adjust R3 CW until a pulse appears at point C.
   c. Adjust R3 CCW until point C just becomes a constant 5.0 ±0.5 Vdc.
   d. Open switch S2 and measure response time t_{RHL}.

FIGURE 24 Response time (t_{RLH} and t_{RHL}) test circuit for table XV - Continued

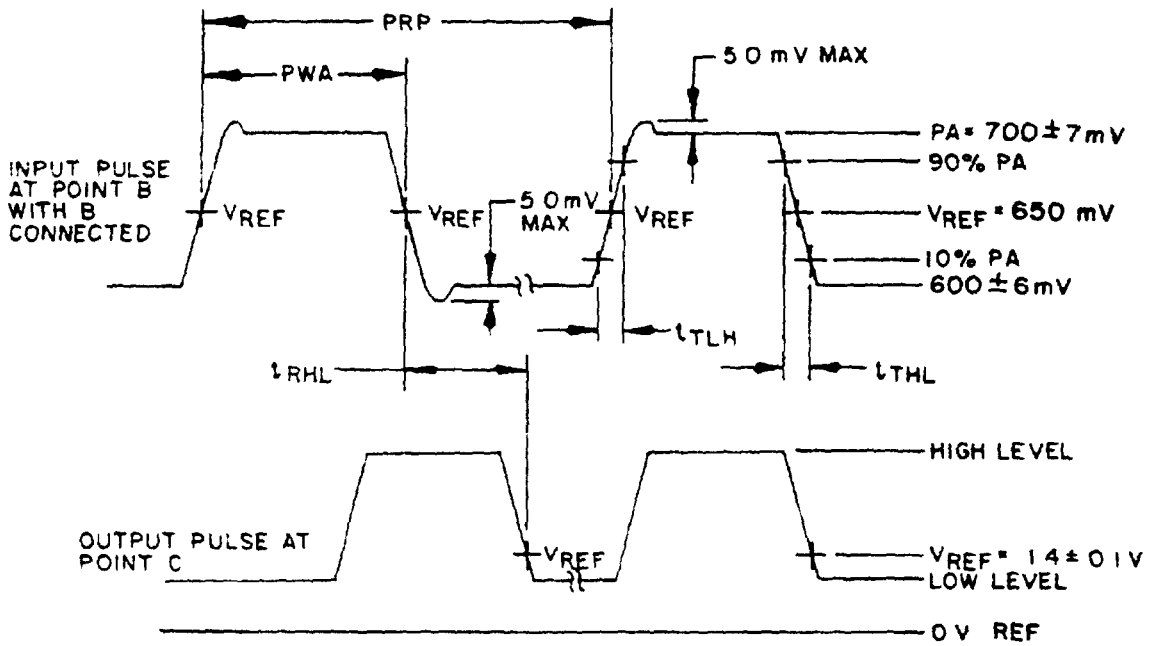
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NOTES
1 Input pulse parameters:
$t_{TLH} = t_{THL} = 5 \pm 1$ ns.
PWA = 500 ±10 ns
PRP = 10 ±1 µs.
2 Before making measurements specified in table XV, verify that the output pulse high level is ≥ 4.5 V and the low level is ≤ 0.6 V
3 All voltages are referenced to ground.

FIGURE 25 Response time (t_{RLH}) test circuit waveforms for table XV (test conditions P1 through P6)

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NOTES
1. Input pulse parameters
   $t_{TLH} = t_{THL} = 5 \pm 1$ ns
   PWA = 500 ±10 ns.
   PRP = 10 ±1 µs.
2. Before making measurements specified in table XV, verify that the output pulse high level is ≥ 4.5 V and the low level is ≤ 0.6 V.
3. All voltages are referenced to ground

FIGURE 26 Response time (t_{RHL}) test circuit waveforms for table XV (test conditions P7 through P12)

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TABLE XVI. High level common mode voltage check.

Test condition	Parameter	Connect	Initial limits 25° +5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
R1	High level common mode voltage	24	3.100	4.545	V	27
R2		3				
R3		23				
R4		13				
R5		12				
R6		31				
R7		24				
R8		3				
R9		23				
R10		13				
R11		12				
R12	High level common mode voltage	31	3.100	4.545	V	27

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